

Class of 2029

Neocortical Development

PPOM1

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**NEW YORK INSTITUTE
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College of Osteopathic
Medicine

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Session Objectives

- Students will be able to describe the basic mechanisms of neuronal proliferation in the fetal ventricular zones and how this relates to neocortical size and disorders such as microcephaly.
- Students will be able to describe the basic principles of neuronal migration in the fetal ventricular zone, subventricular zone, and cortical plate and how this relates to the establishment of the neocortical lamina.
- Students will be able to describe the mechanisms underlying disorders of neuronal migration such as subcortical band heterotopia, lissencephaly, etc.
- Students will be able to describe the basic organization of the formation of sulci and gyrii in the developing neocortex.



3P's of Brain Development & Function

Neuronal Production – **proliferation** of neurons in order to appropriately populate the brain.

Neuronal Placement – the **migration** of neurons to their appropriate laminar and areal position.

Neuronal Performance – intrinsic neuronal **physiology** (ion channels & transmitter receptors), synaptic physiology of networks of neurons

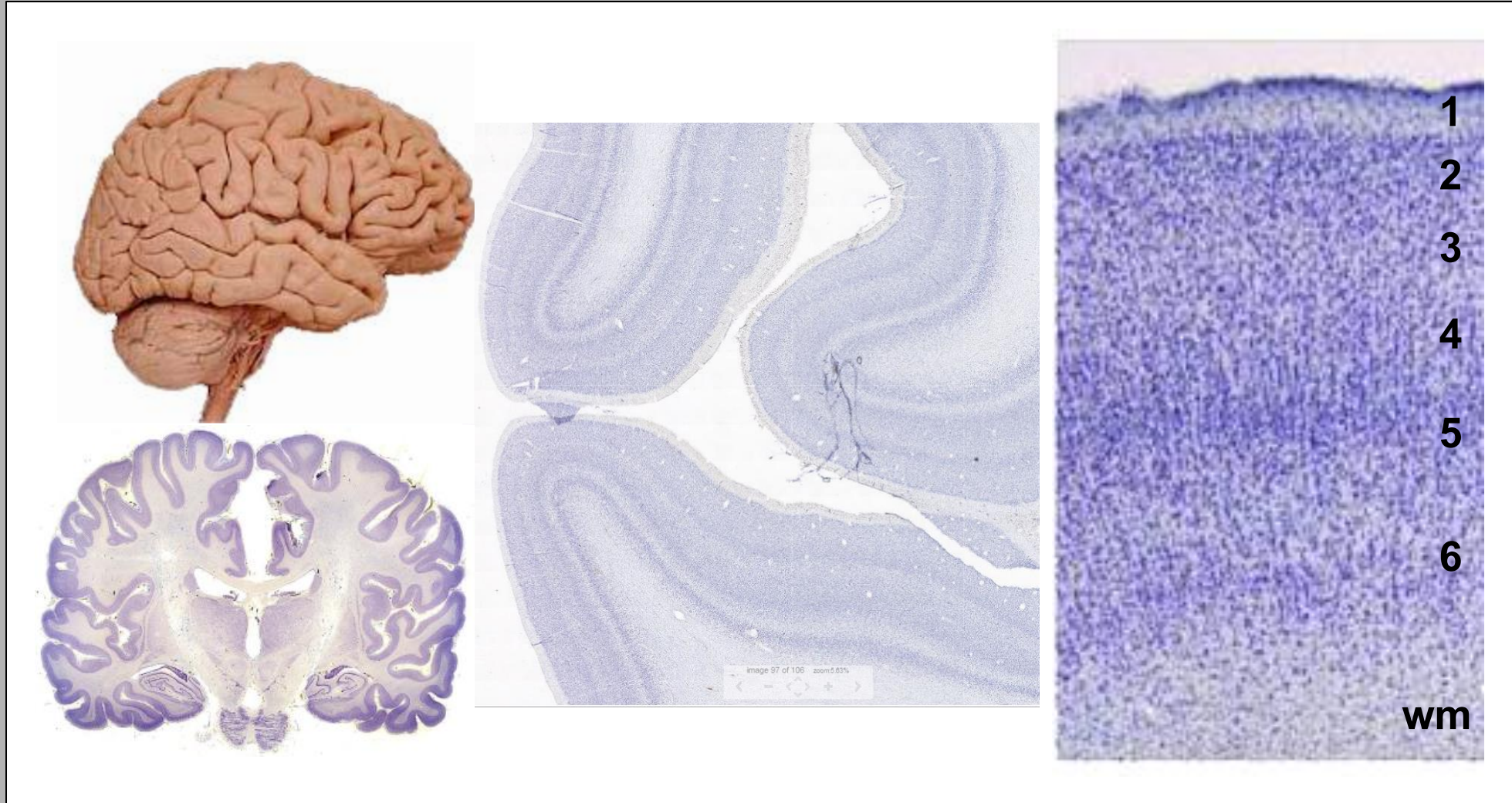
***THE 3P's ORGANIZE AND CONTROL BEHAVIOR:
sensation & perception, cognition & memory, motor
control, sleep/wake, emotion, etc.***



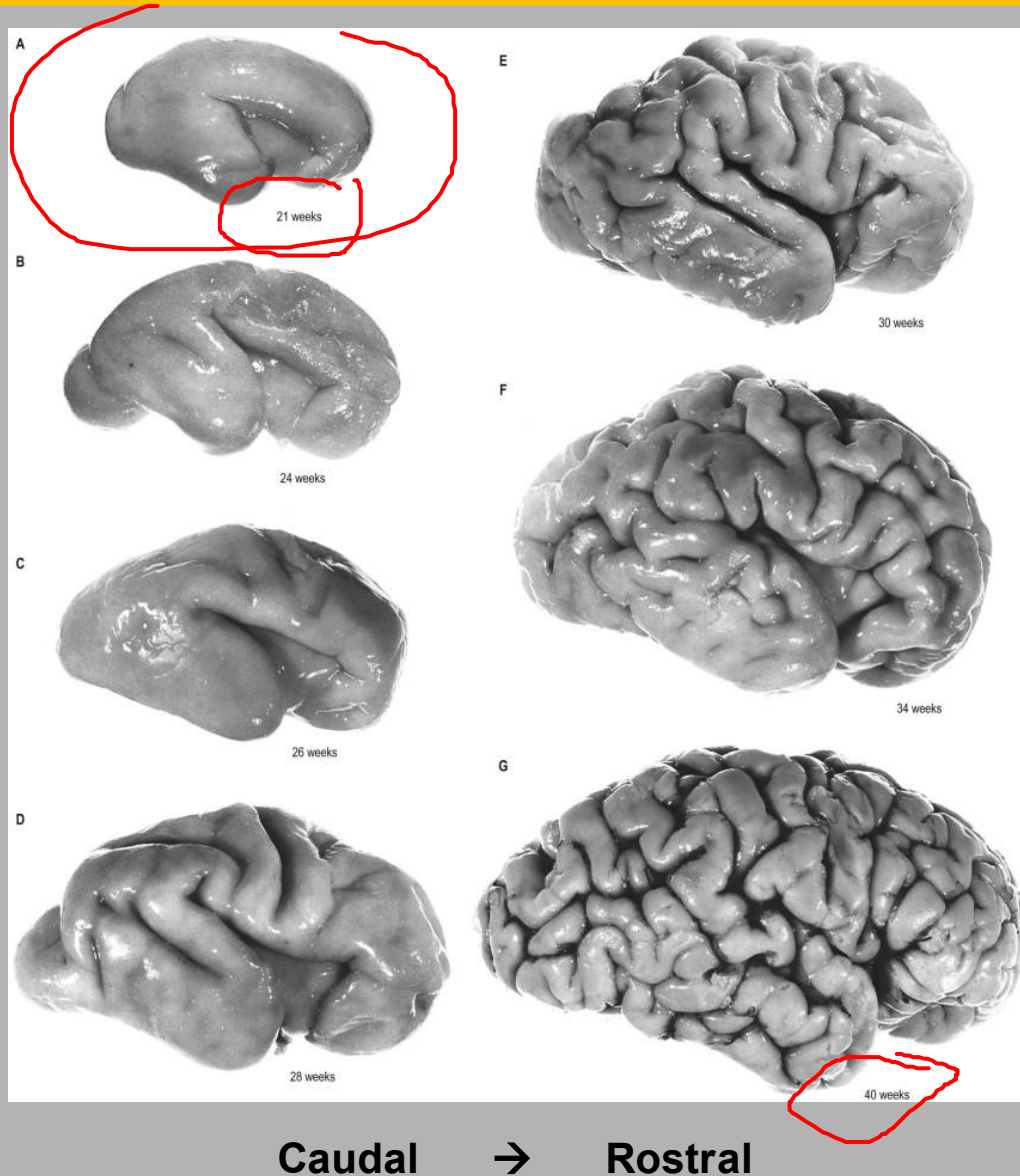
Neuronal Production

- making and keeping the right #'s of neurons

The adult mammalian neocortex displays a laminated cytoarchitecture



3P's of Brain Development & Function



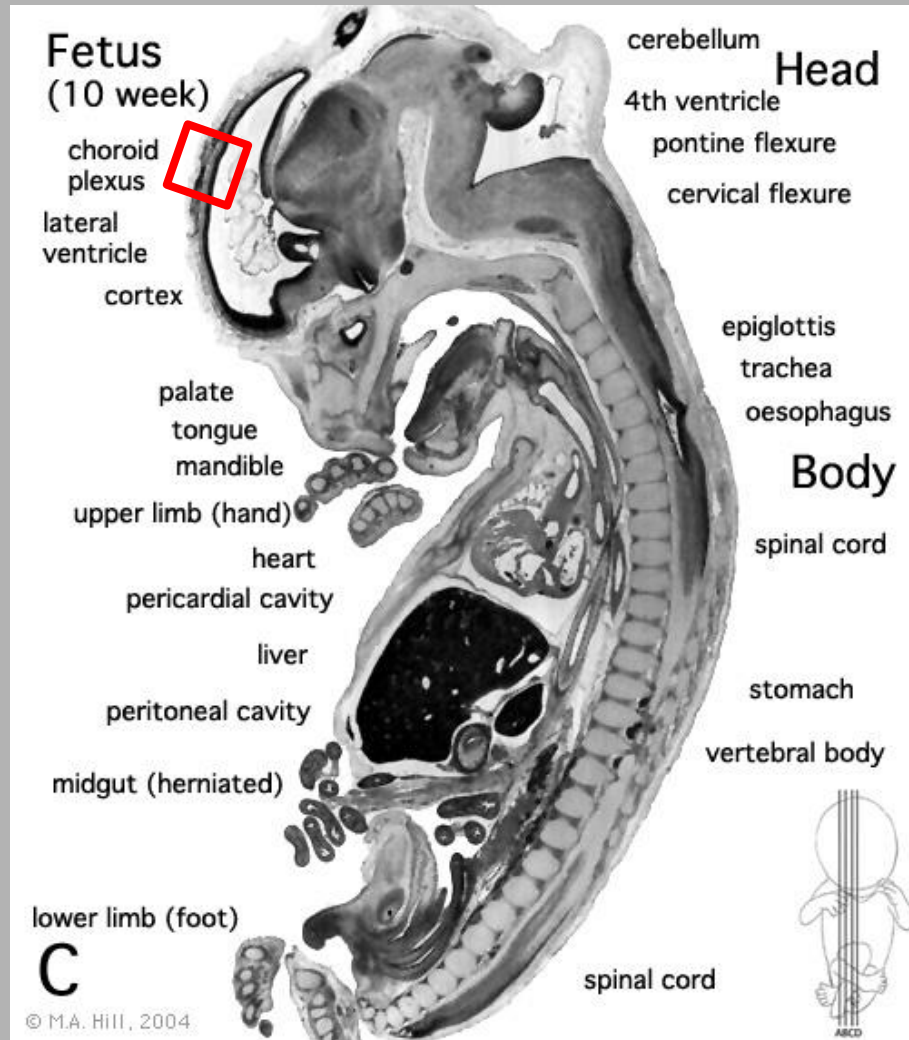
Brain is growing in size indicative of cellular proliferation.

Neocortex is changing from smooth surface to full of sulci and gyri – including development of lamina (layers).

Lamina are found in many tissues/organs.

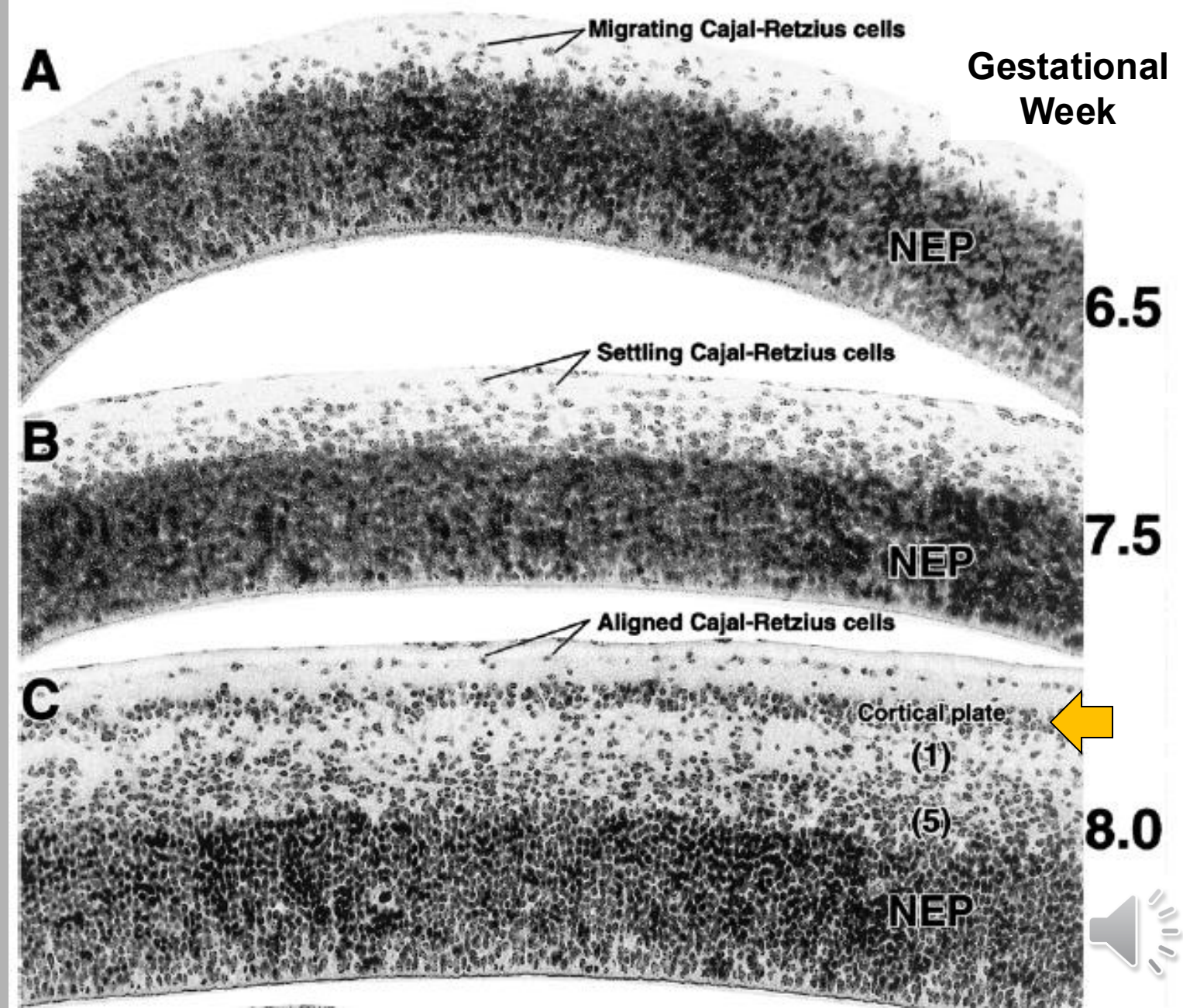


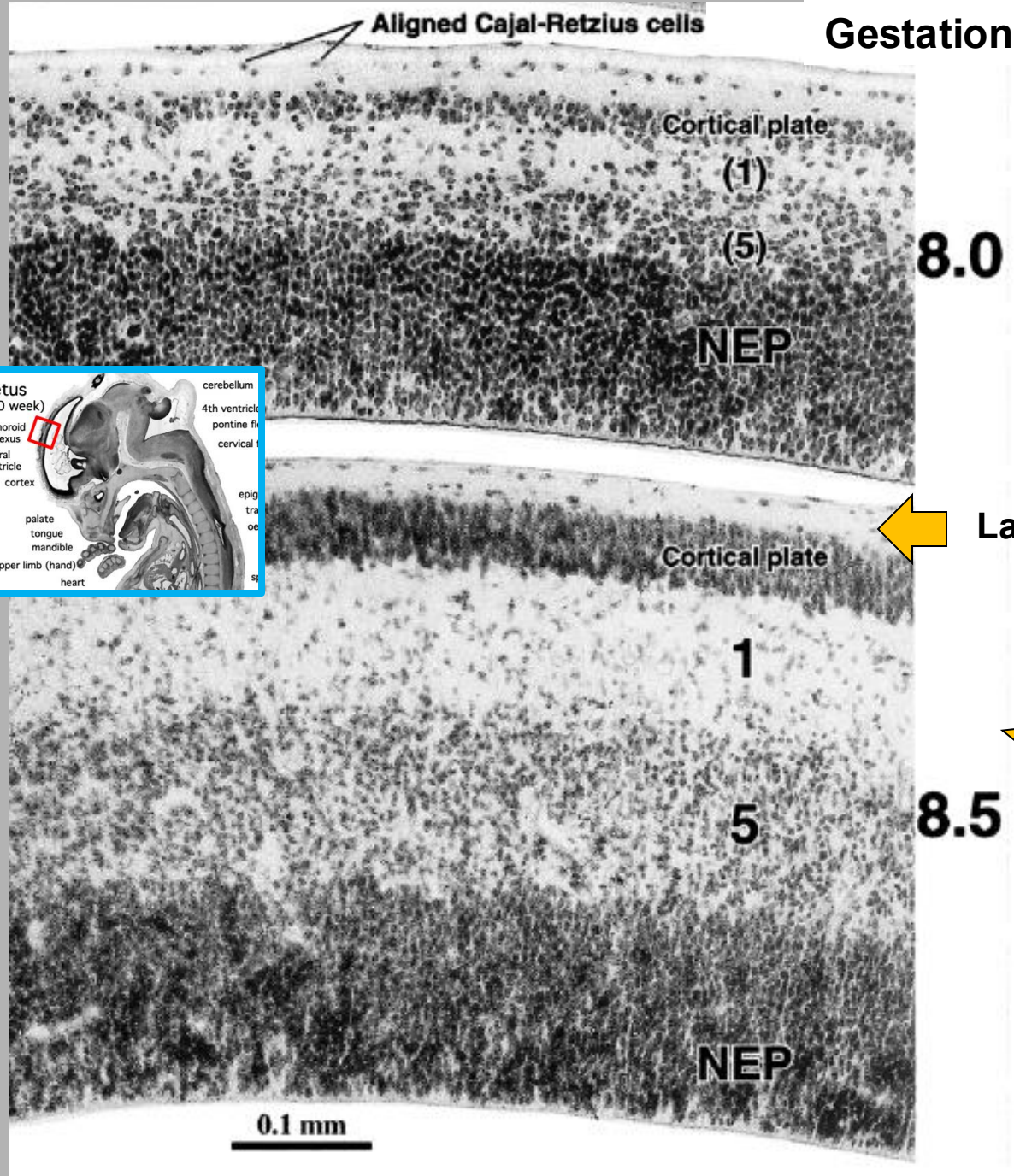
Neuronal Production - making and keeping the right #'s of neurons



Neocortical development

- nearly all cortical neurons are made prenatally
- all excitatory neurons are made at the VZ

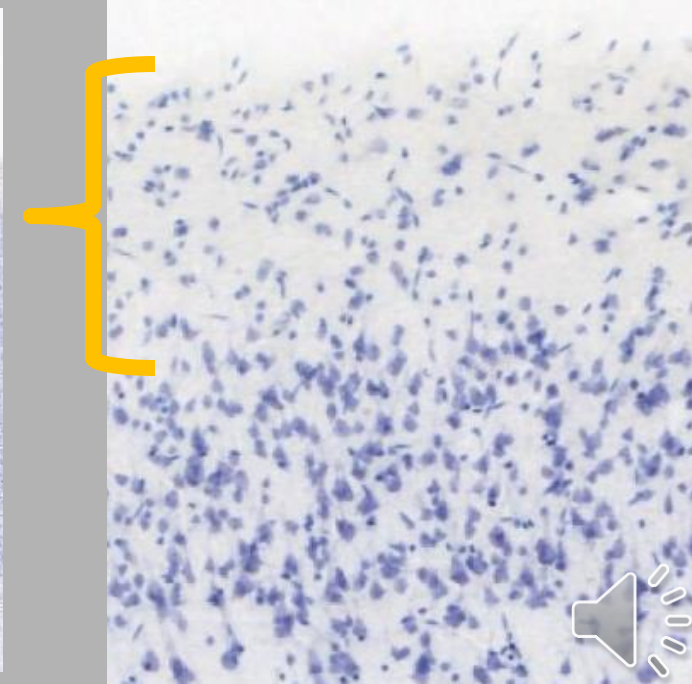
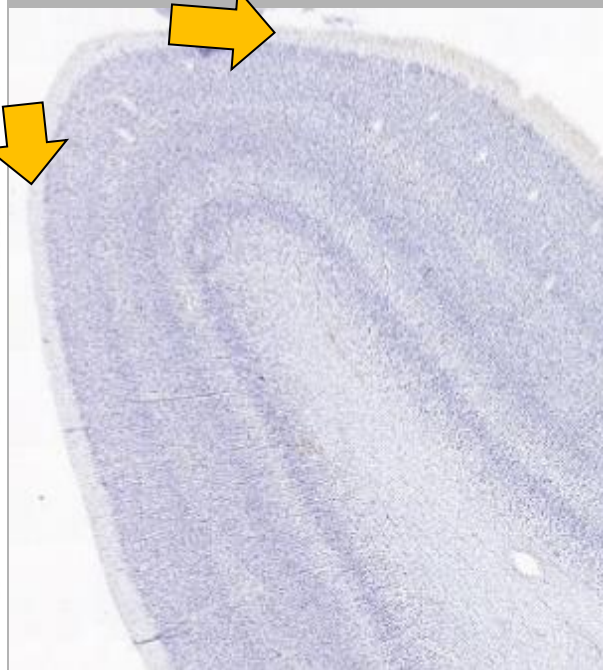




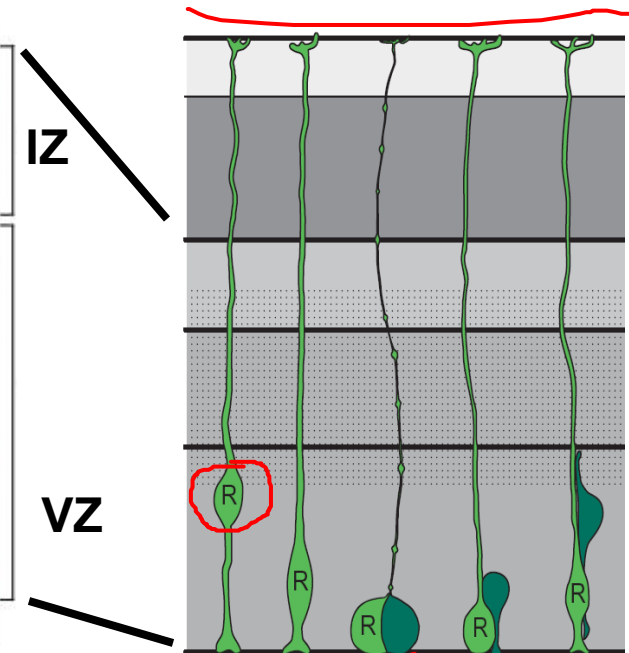
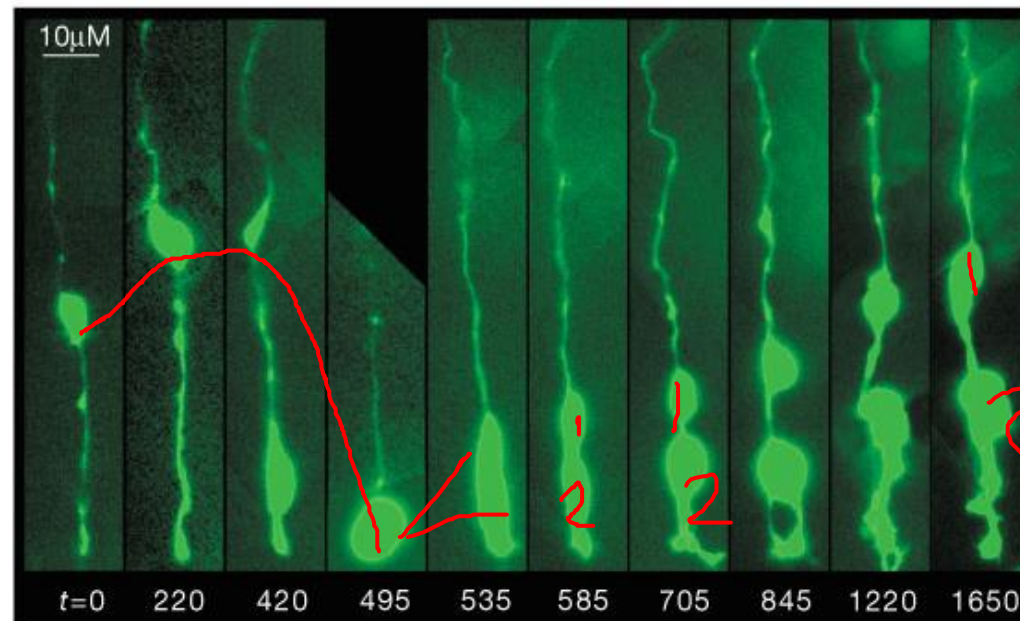
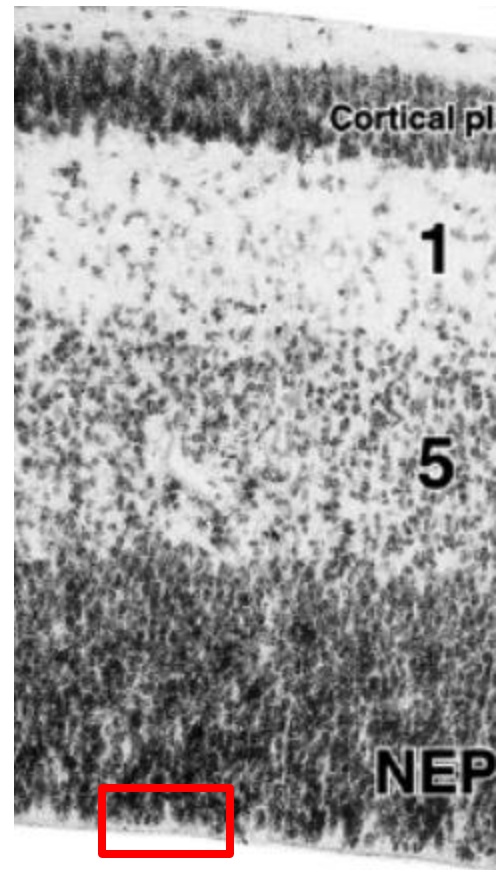
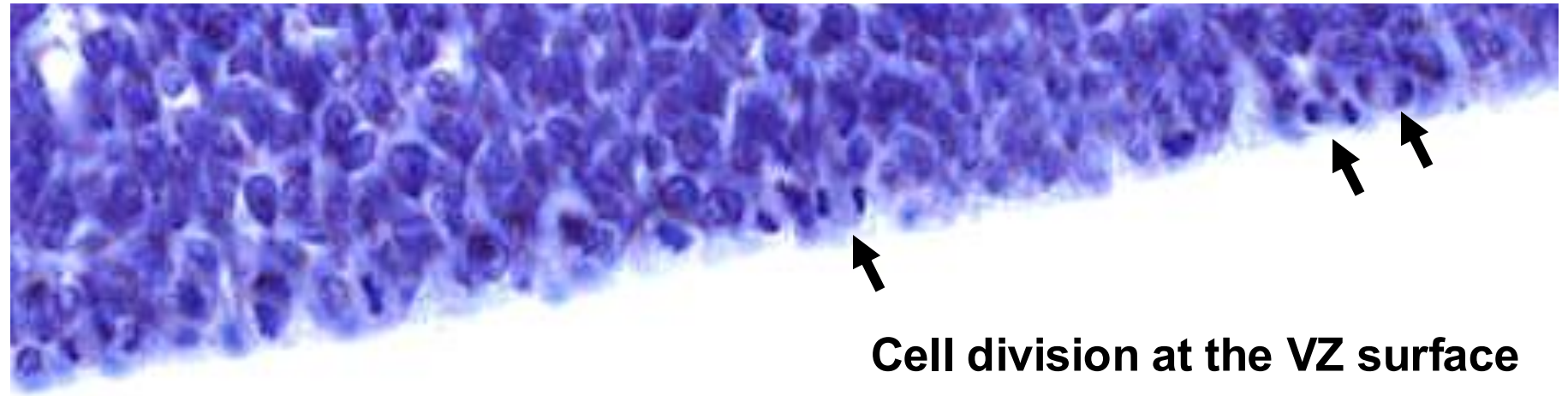
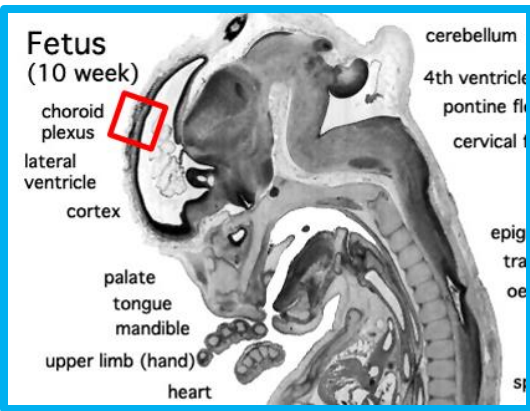
Gestational Week

- More neurons being produced by neuroepithelial cells (radial glia) at VZ
- Newborn neurons migrate to CP which is getting thicker.
- Layer 1 not getting thicker.

Layer 1/ Marginal zone / Molecular layer



Radial glial cells divide to produce neurons which become the pyramidal neurons that populate the neocortex



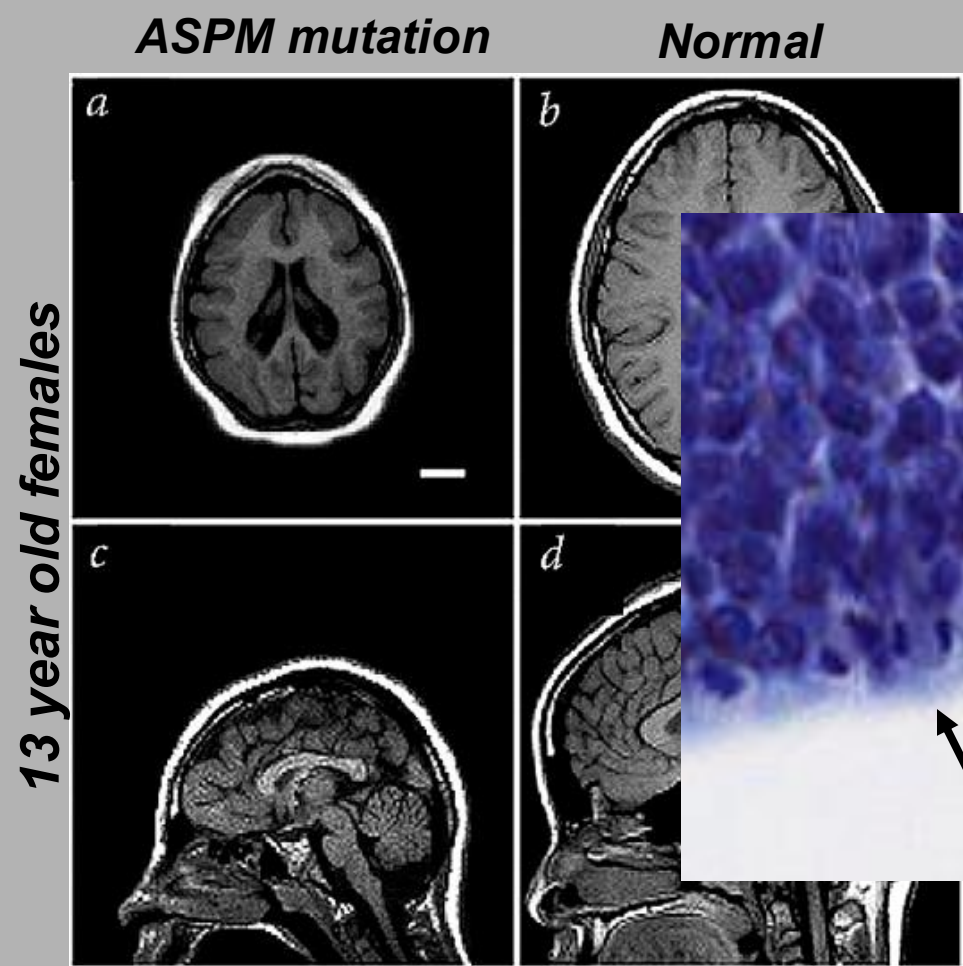
Noctor et al. 2001

Kriegstein & Noctor 2002

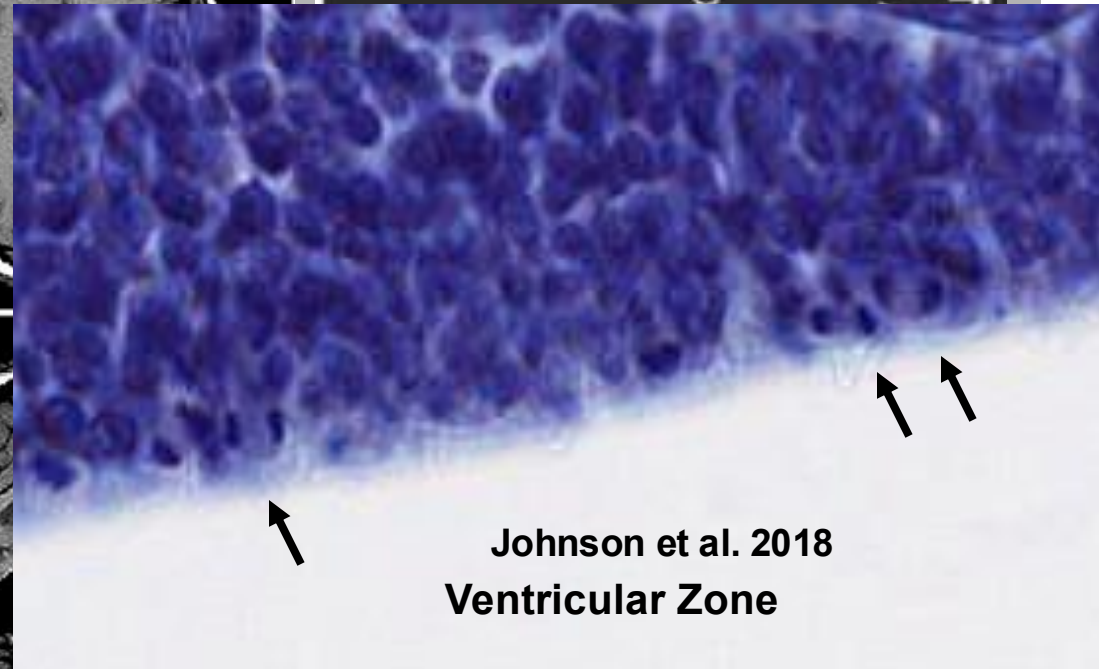


Why is it important to understand neuronal proliferation in the brain?

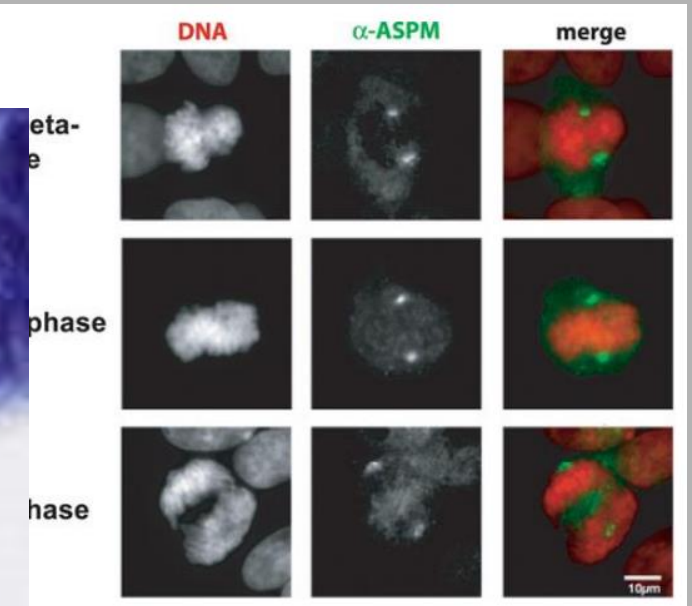
Neuronal production disorders: *primary microcephaly*



Bond et al. 2002



* **ASPM** important for mitotic spindle formation



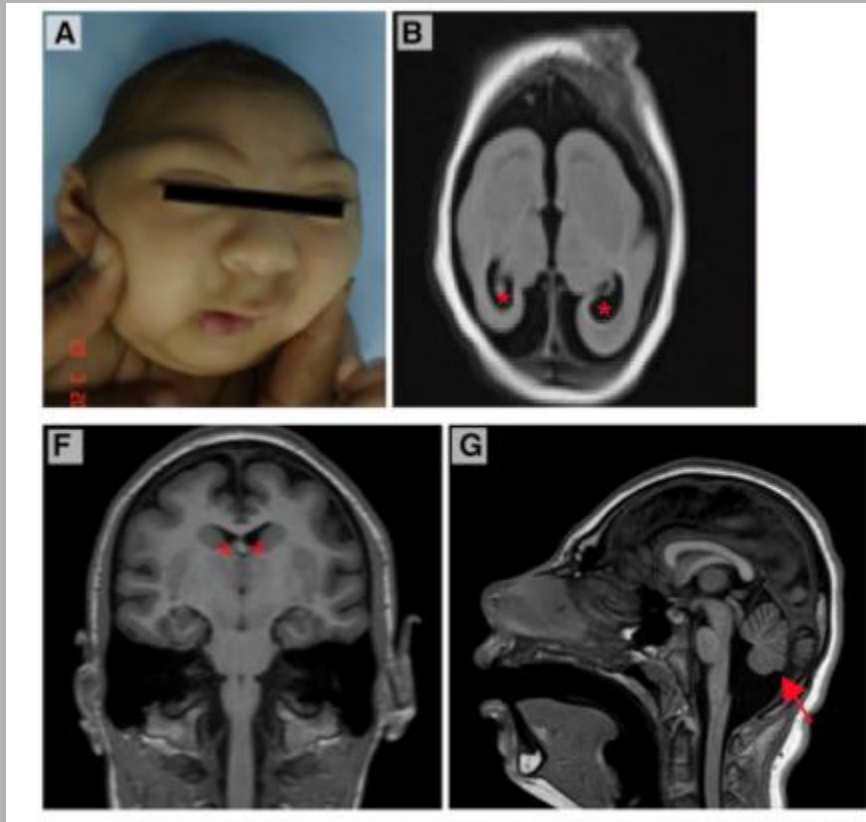
Kouprina et al. 2005



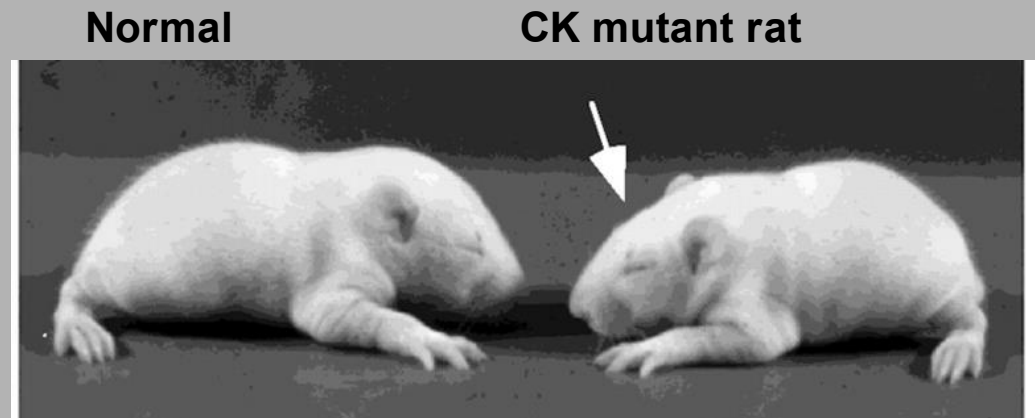
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Neuronal production disorders: *primary microcephaly*

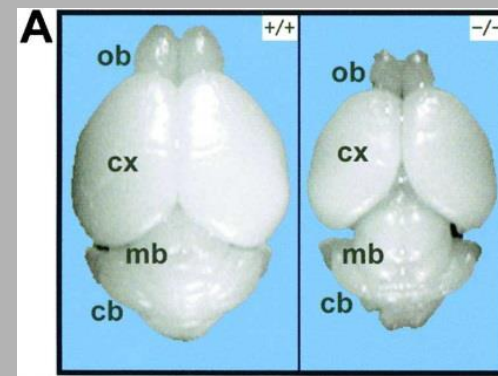
- * Citron kinase (CK) mutations cause microcephaly in humans, rats, and mice



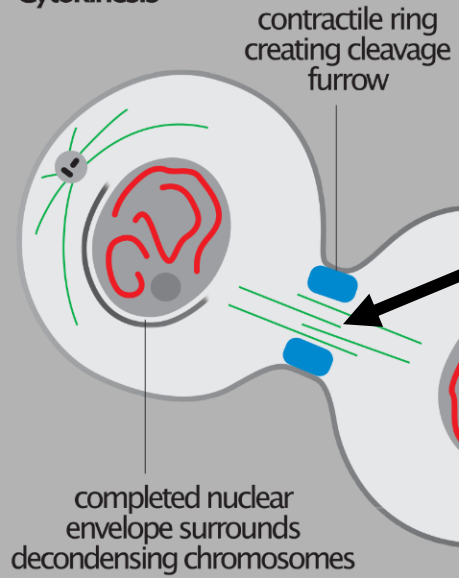
Harding BN, et al. Am J Hum Genet. 2016 Aug 4;99(2):511-20.
Roberts et al. 2000; J Neuroscience
DiCunzio et al. 2000; Neuron



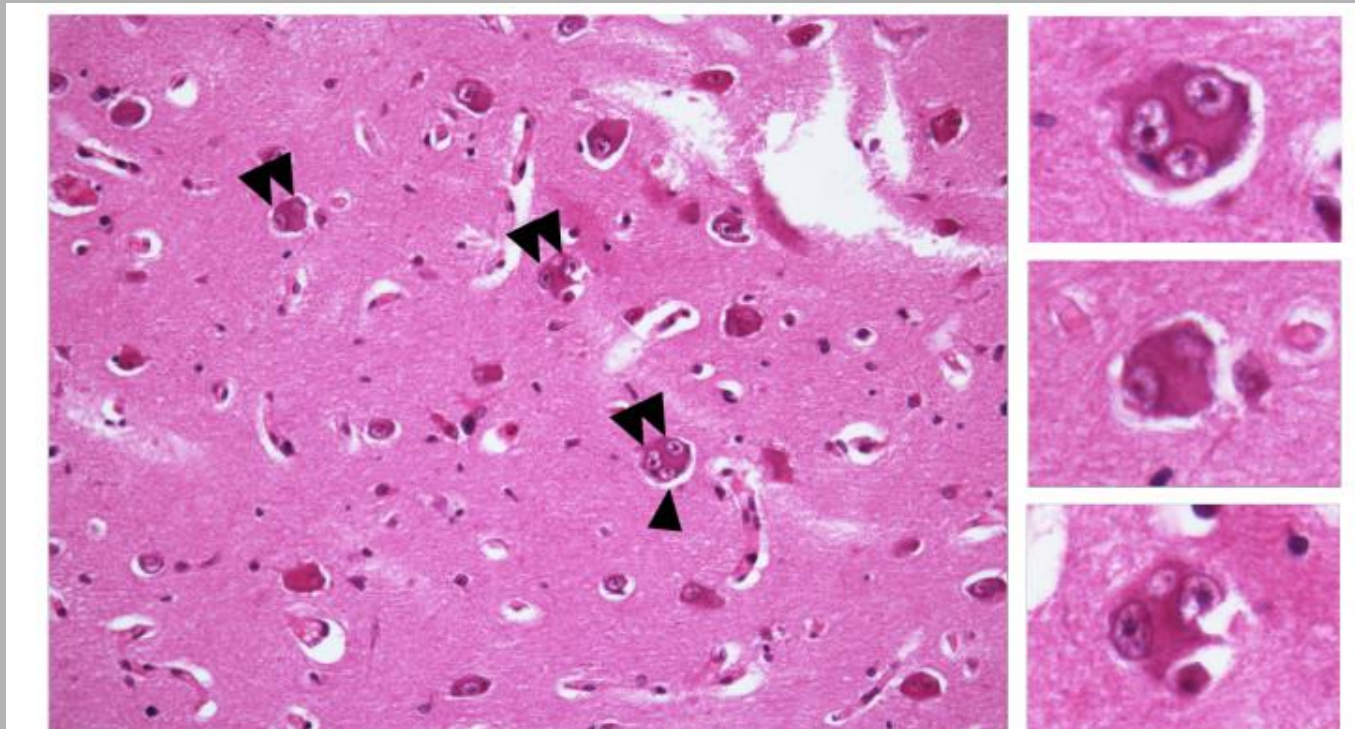
Normal CK KO mouse



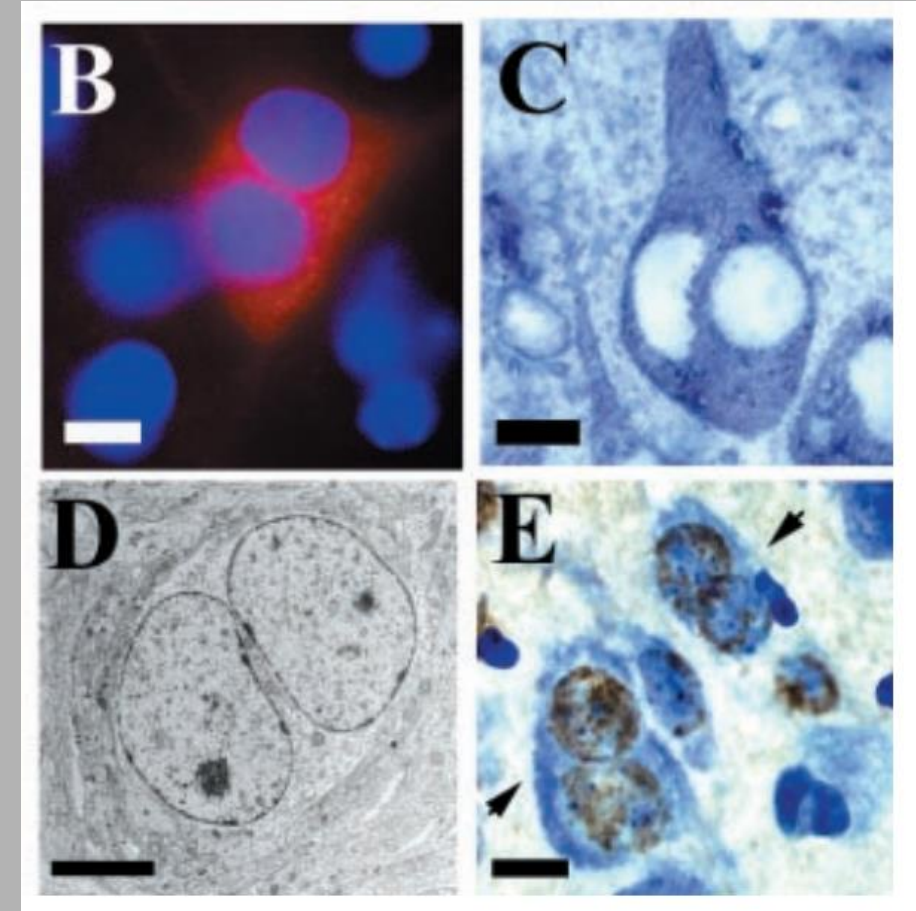
Cytokinesis



Citron Kinase mutations cause microcephaly by impairing cytokinesis in radial glia resulting in binucleate neurons and a reduction in the total number of neurons in brain.



Harding BN, et al. Am J Hum Genet. 2016 Aug 4;99(2):511-20.



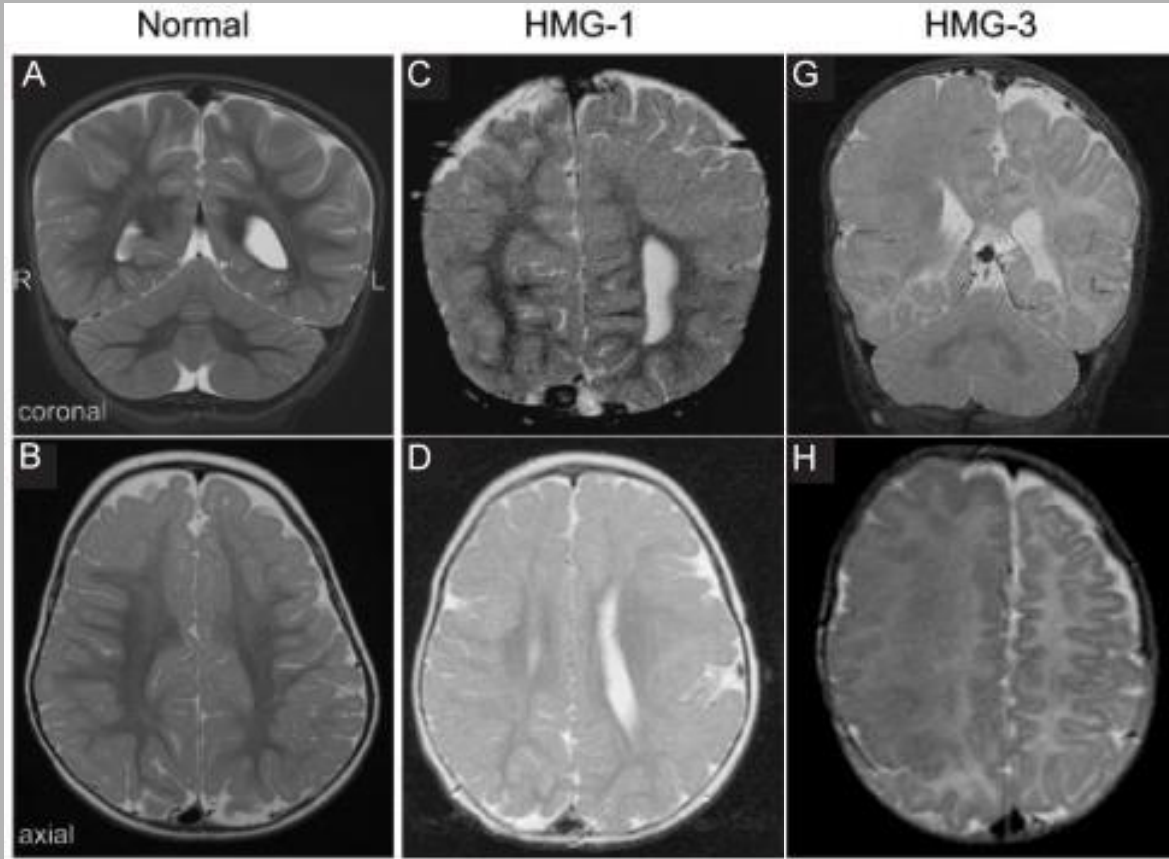
Sarkisian et al. 2002; J Neuroscience



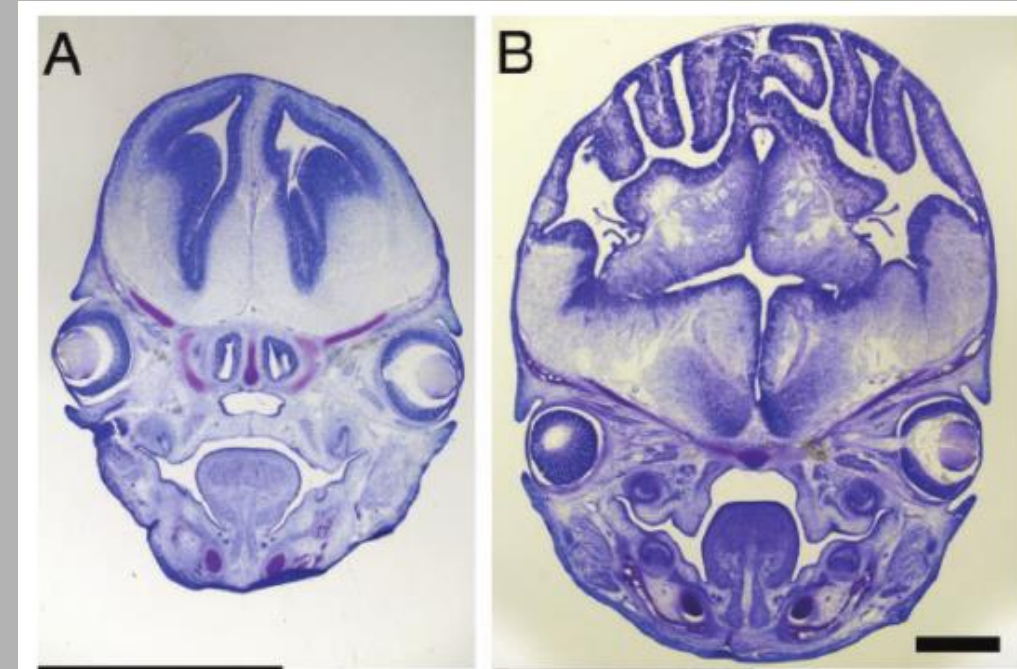
Why is it important to understand neuronal proliferation in the brain?

Neuronal production disorders: *Hemi-megalecephaly*

Poduri et al. Neuron. 2012 Apr 12; 74(1): 41–48.



Experimental megalecephaly by increasing proliferation in radial glia.



Beta catenin /nestin over exp;
Chenn A & Walsh CA 2002

* Enlarged and malformed hemisphere with associated epilepsy and cognitive delay/disability
Genetic studies show - activating mutation in *AKT3* which causes prolonged proliferative capacity of radial glia. *AKT3* expression found in radial glia at VZ



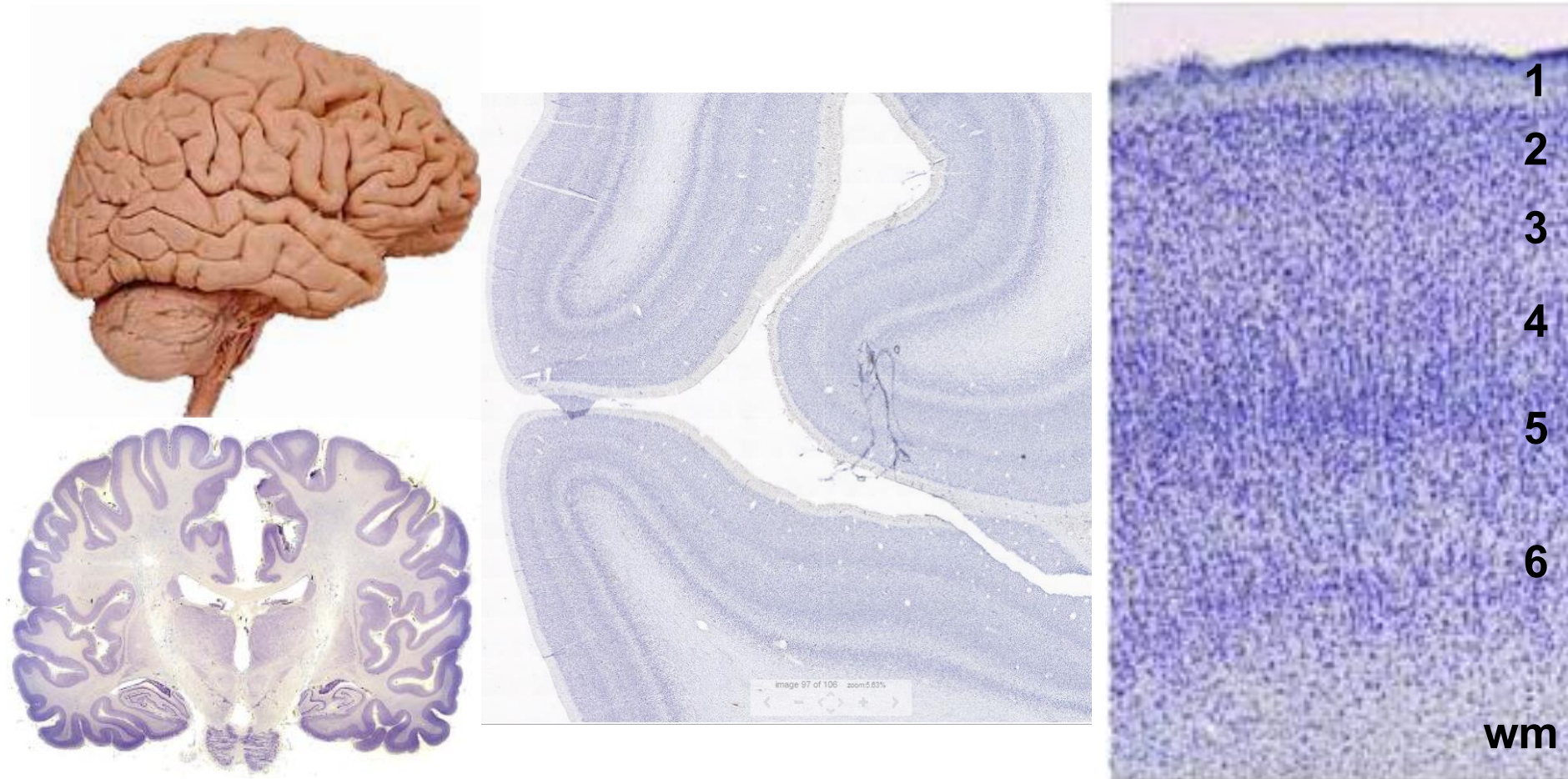
- Neuronal Production – proliferation of neurons in order to appropriately populate the brain.
- Radial glial are the proliferative cells that divide to make neurons at the ventricular zone of the neuro-epithelium
- Too much or too little neuronal production leads to microcephaly or megalencephaly



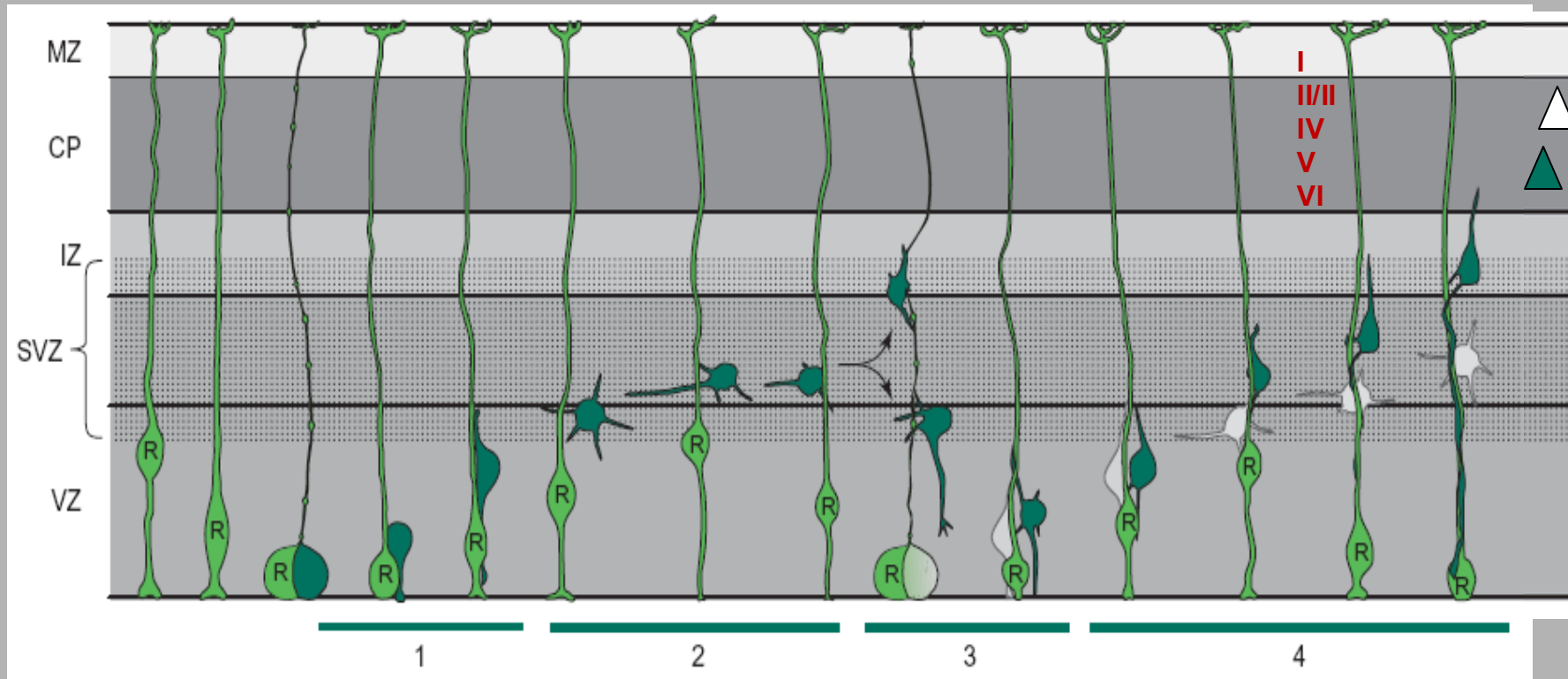
Neuronal Placement in the developing neocortex

- establishing neuronal lamina

The adult mammalian neocortex displays a laminated cytoarchitecture



Neuronal placement includes dynamic cellular movement and dramatic changes in morphology



SHAPE: Bipolar
MVMNT: Radial

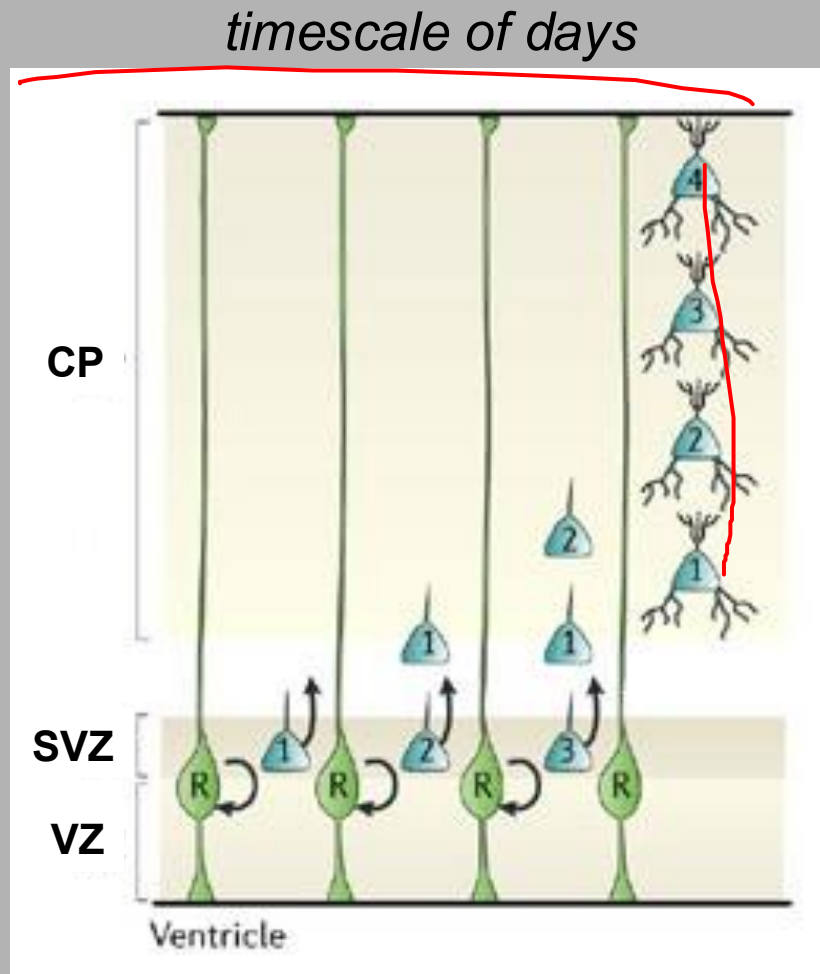
SHAPE: Multipolar
MVMNT: Dynamic

SHAPE: Bipolar
MVMNT: Radial

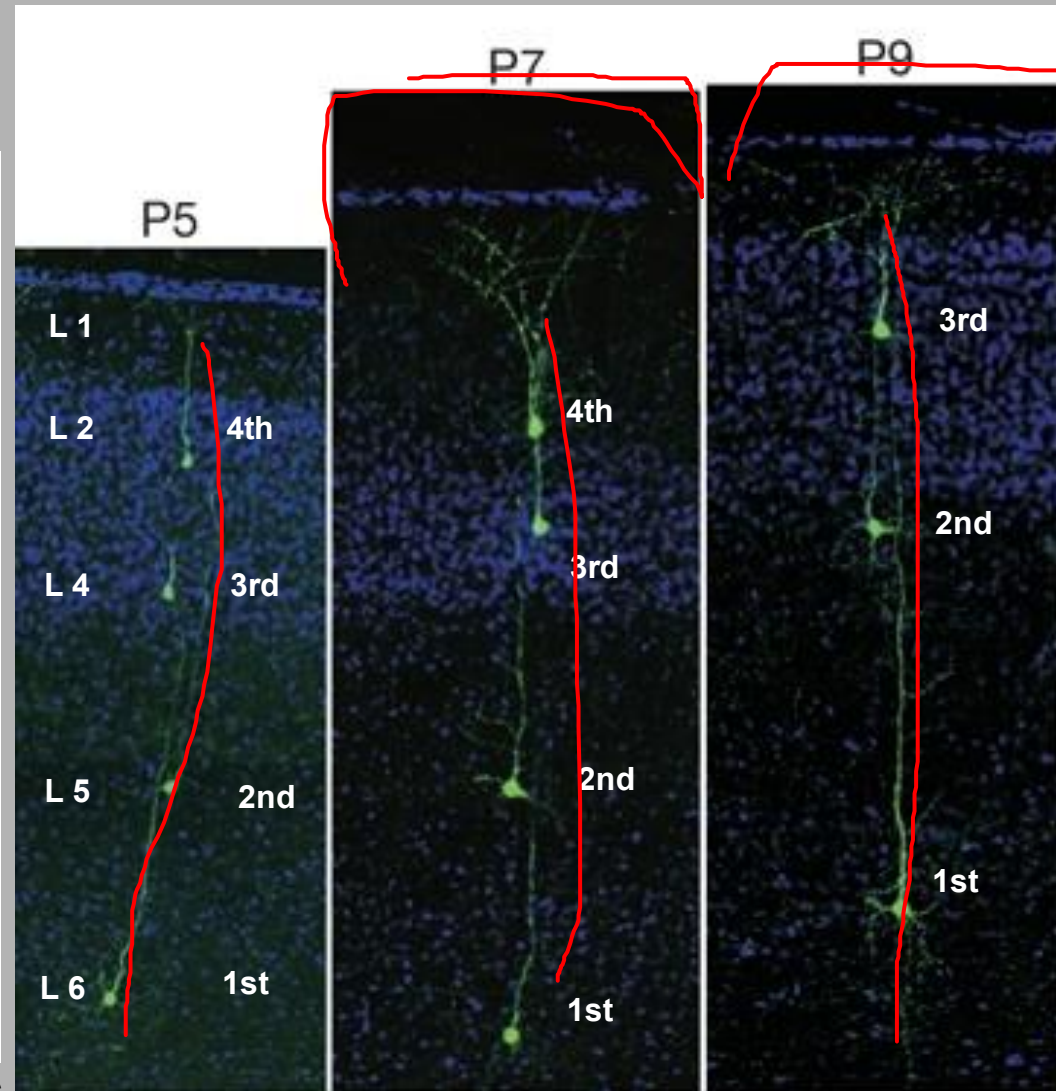
SHAPE: Bipolar
MVMNT: Radial



Sequential neuronal proliferation and inside-out radial migration establishes neocortical lamina



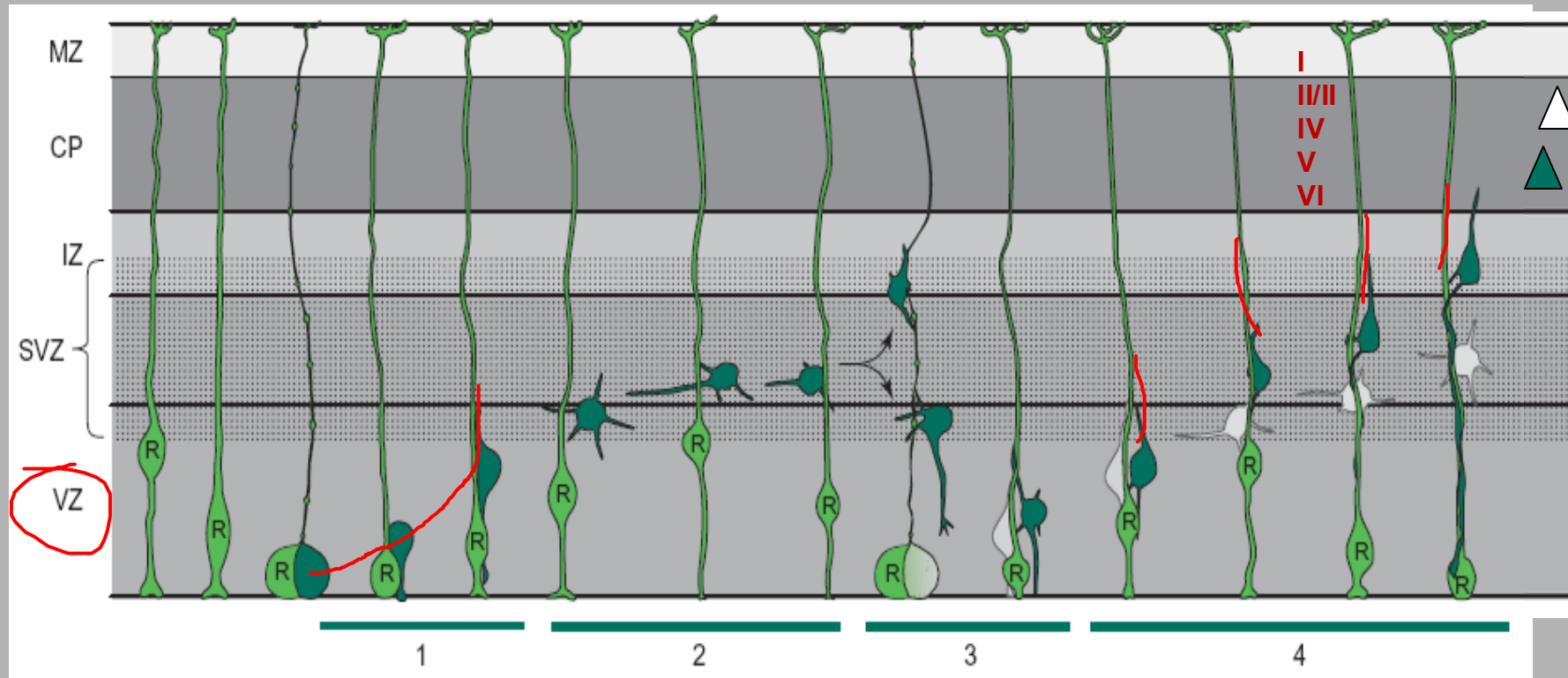
Kriegstein et al. 2006



Nature 458, 501-504 (2009)



Neuronal placement includes dynamic cellular movement and dramatic changes in morphology



SHAPE: Bipolar
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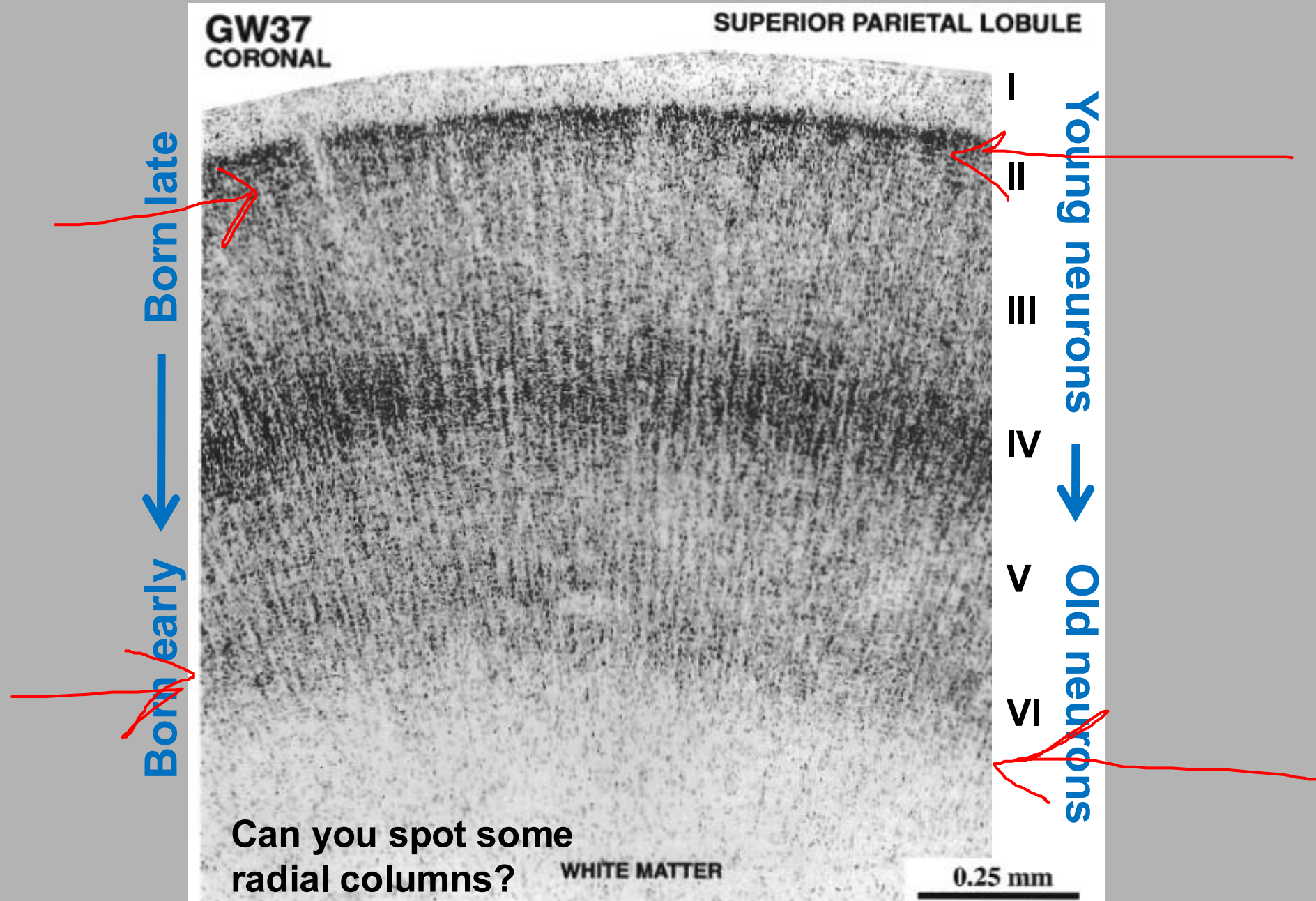
SHAPE: Multipolar
MVMNT: Dynamic

SHAPE: Bipolar
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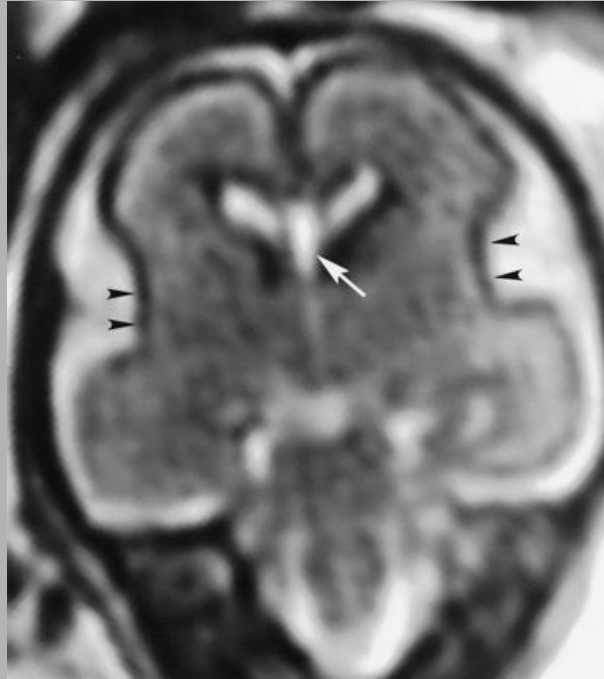


Inside-out radial migration establishes neocortical lamination

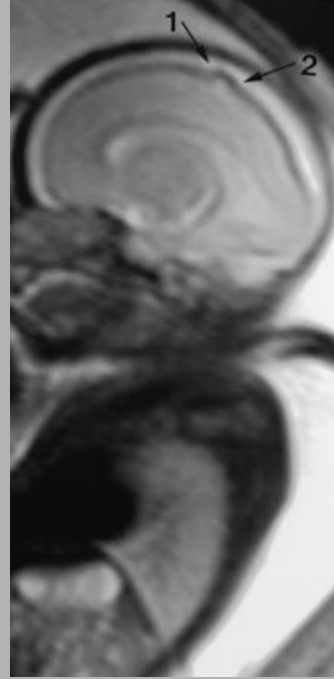


Development of neocortical sulci & gyri

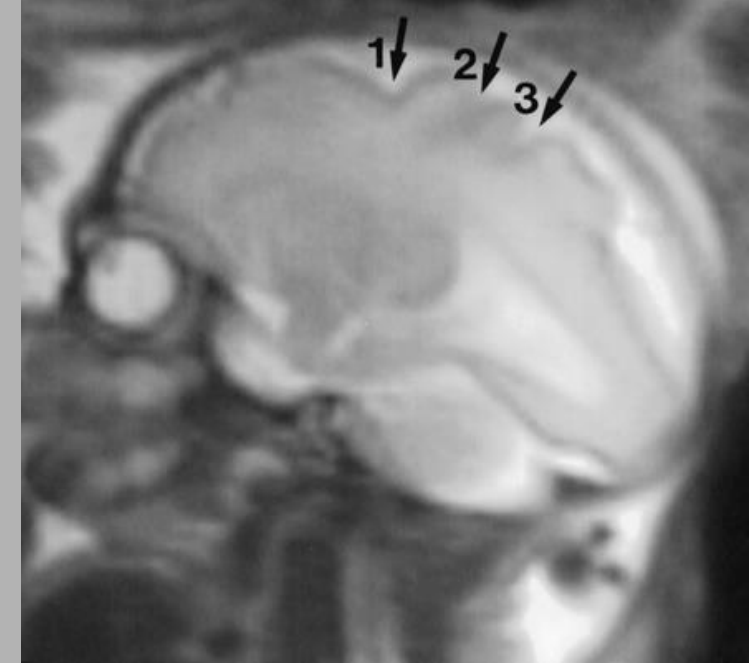
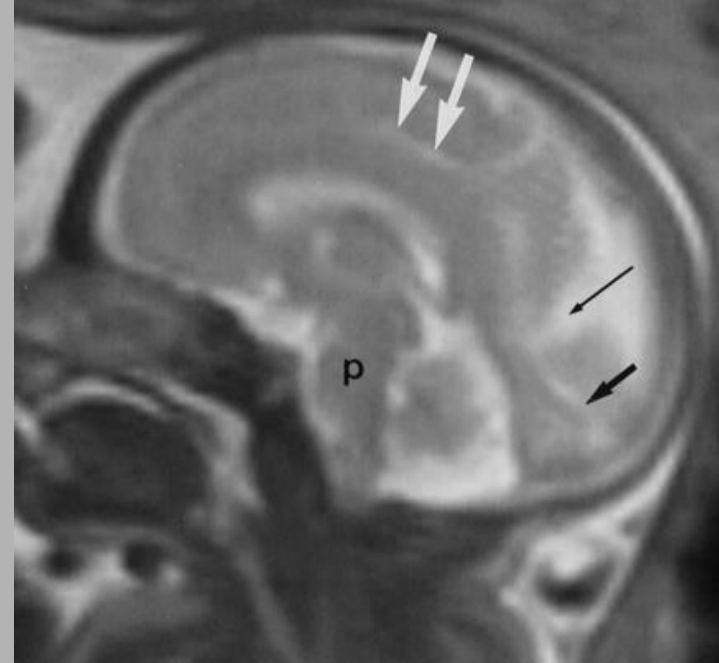
18 WEEKS



26 WEEKS



30 WEEKS



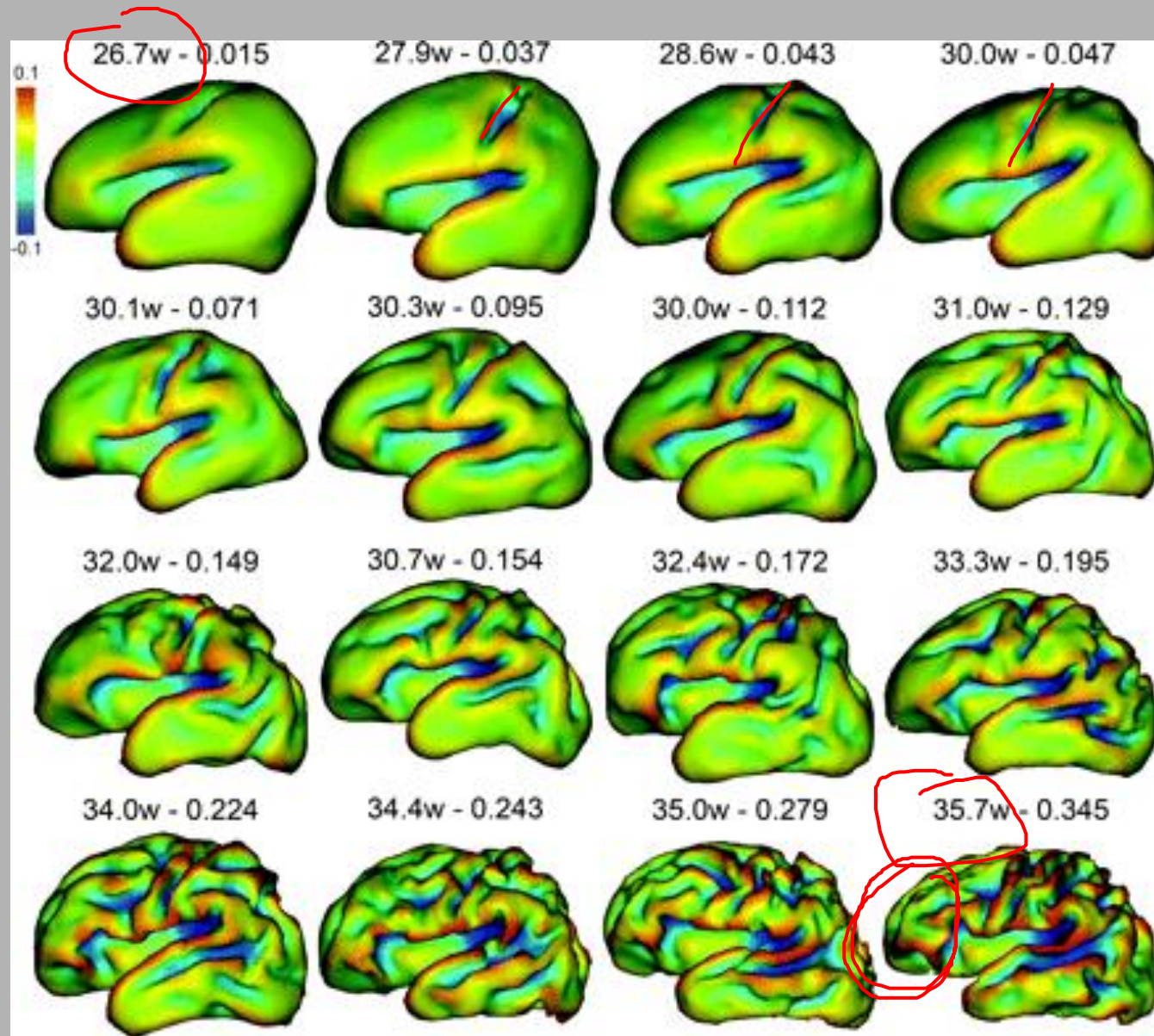
The sylvian fissure begins as a shallow depression at 14 weeks gestation and subsequently becomes grooved.

Early appearance of the precentral (arrow 1) and central (arrow 2) sulci, is depicted.

Left, The cingulate sulcus (white arrows), parieto-occipital fissure (thin black arrow), and calcarine fissure (thick black arrow) are seen. *p* = pons. Right, the precentral (arrow 1), central (arrow 2), and postcentral (arrow 3) sulci are seen.



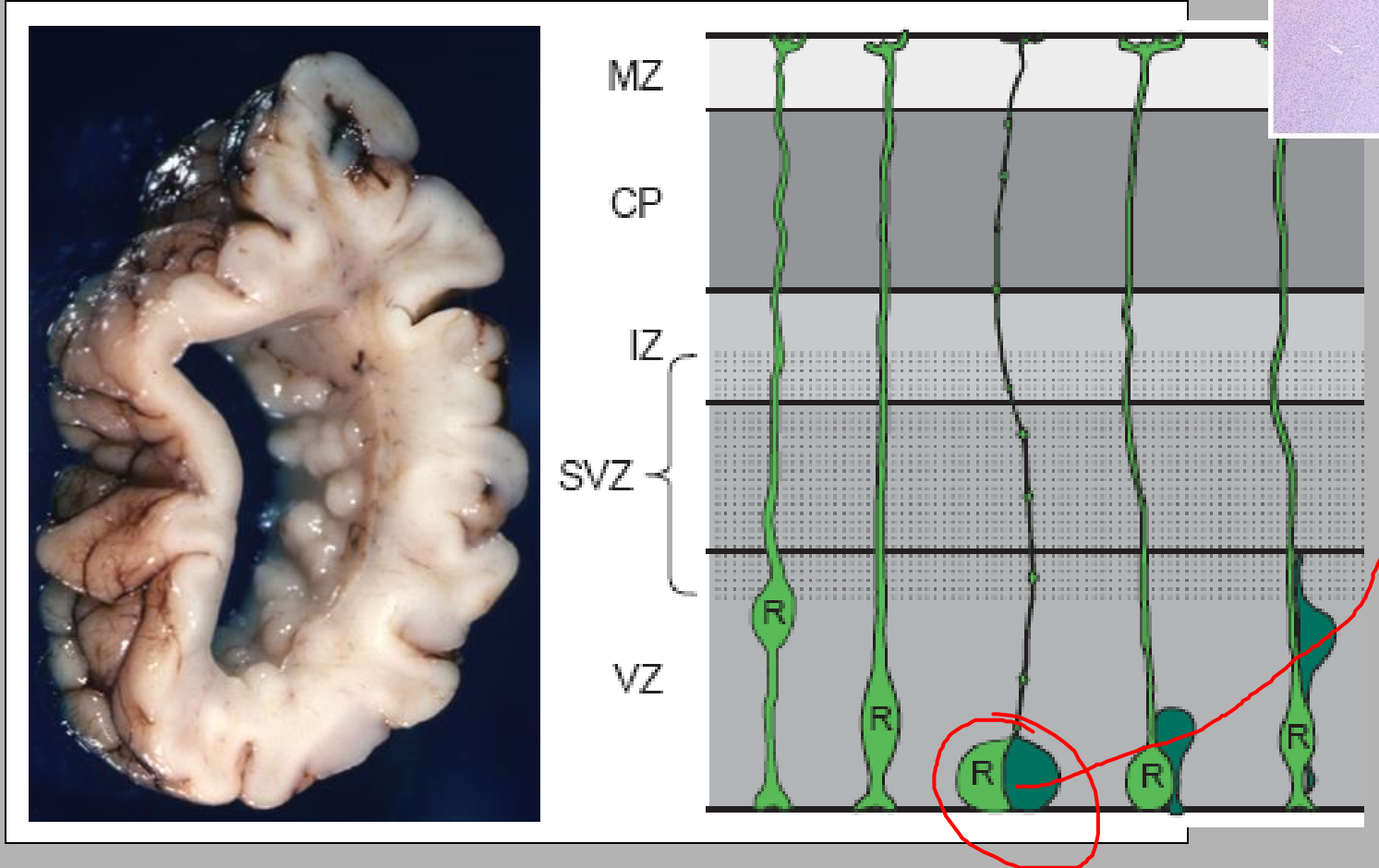
Large fissures are the first to develop. Frontal lobe sulci/gyri are the last to develop.



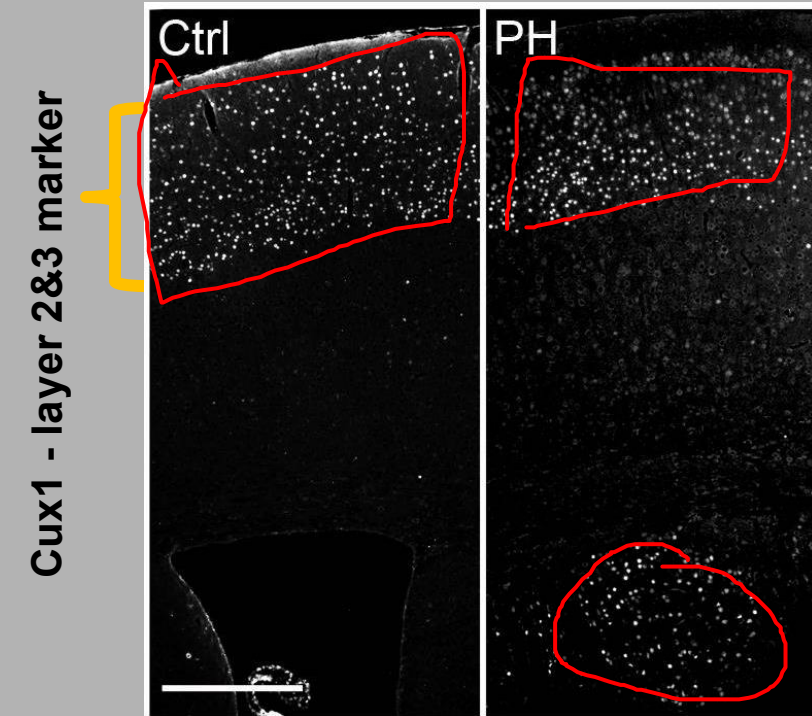
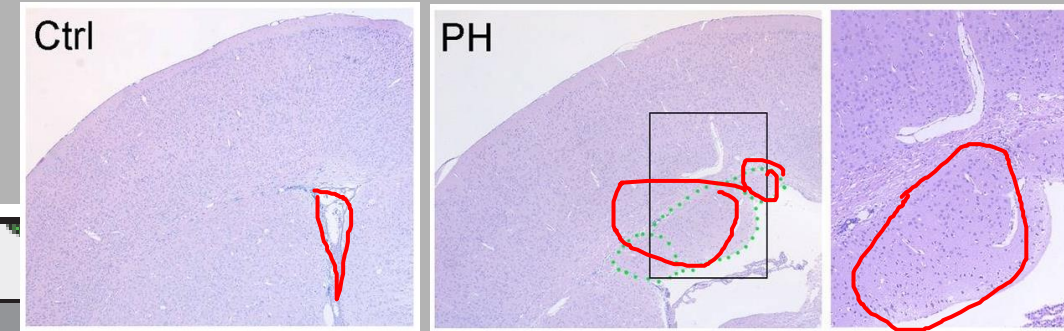
Neuronal placement disorders:

Periventricular-nodular heterotopia (PVH)

- neurons fail to leave the ventricular zone forming nodules



PVH in mice w/ Flna mutation



- associated with mutations of Filamin A (*Flna*) gene – actin binding protein
- accompanied by cognitive impairment and epilepsy

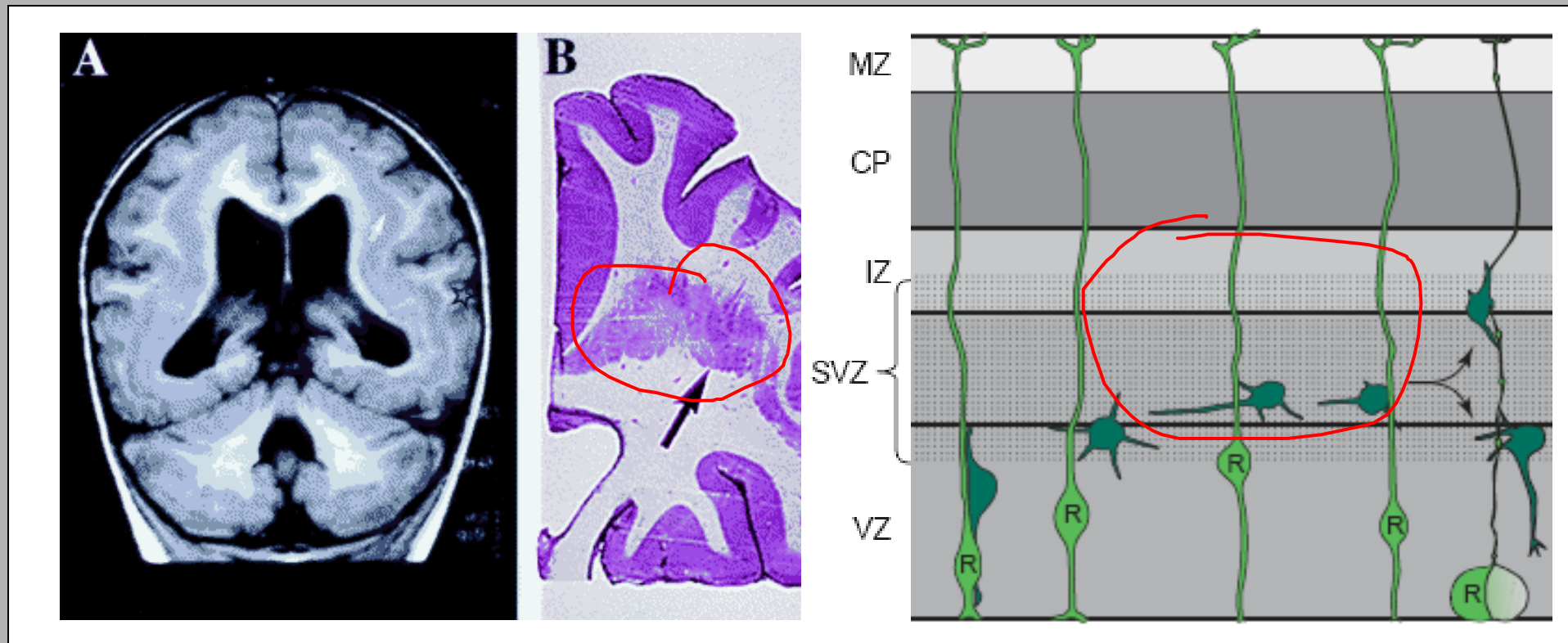
Houlihan SL et al. 2016



Neuronal placement disorders:

Subcortical Band Heterotopia/ “Double-cortex syndrome”

- *neurons stuck in the IZ which becomes the white matter. Heterotopia are surrounded by WM*
- *associated with mutations of the EML1, DCX, LIS1 – all microtubule associated proteins*
- *accompanied by cognitive impairment and epilepsy*

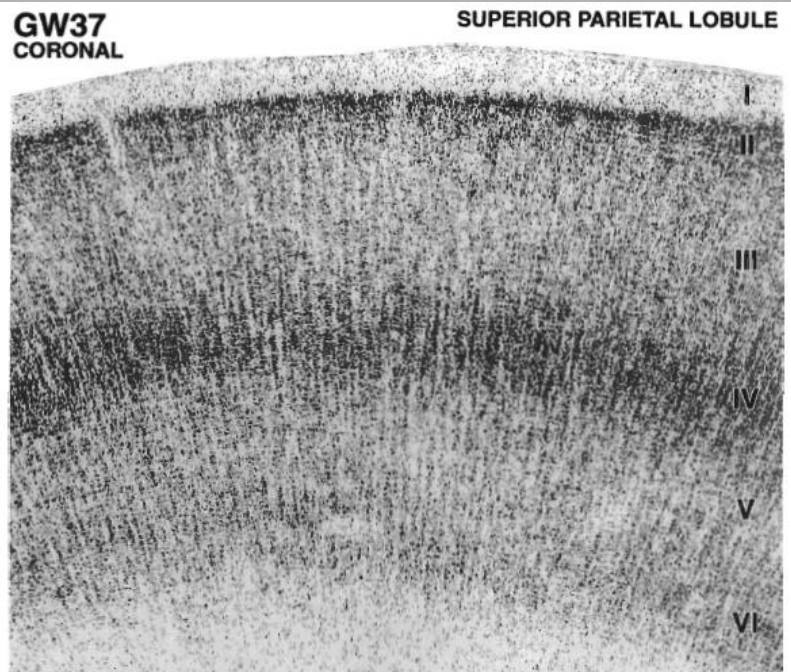


- *neurons fail to leave the intermediate zone (which becomes the WM)*



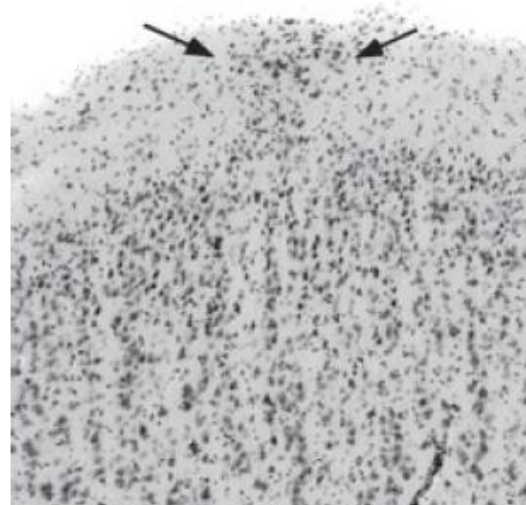
Neuronal placement deficits can result in *Molecular-layer heterotopia*

- *neurons migrate too far into the molecular layer*

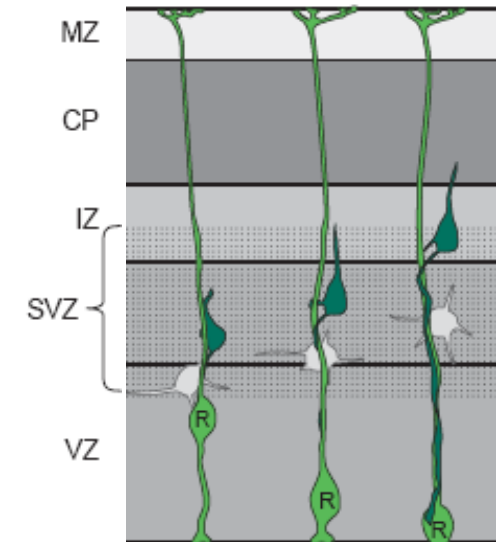


Molecular Layer
aka Layer I

Layer 2/3



Geschwind 1990



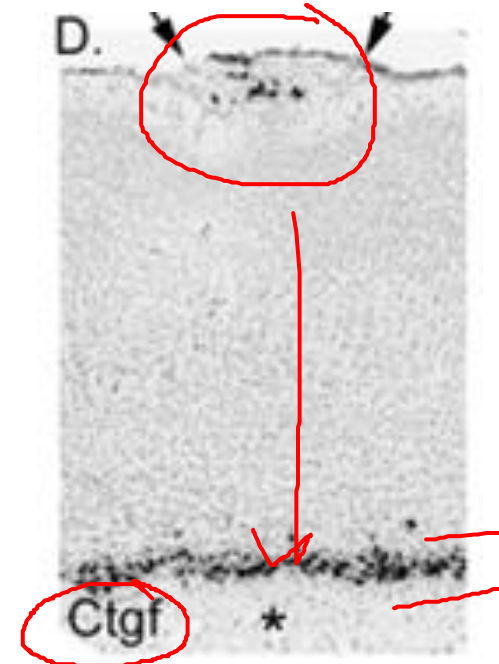
Found in brains of epilepsy patients and dyslexic patients.



Neuronal placement deficits can result in *Molecular-layer heterotopia*

Misplaced neurons belonging in all layers are found in molecular layer heterotopia in mice.

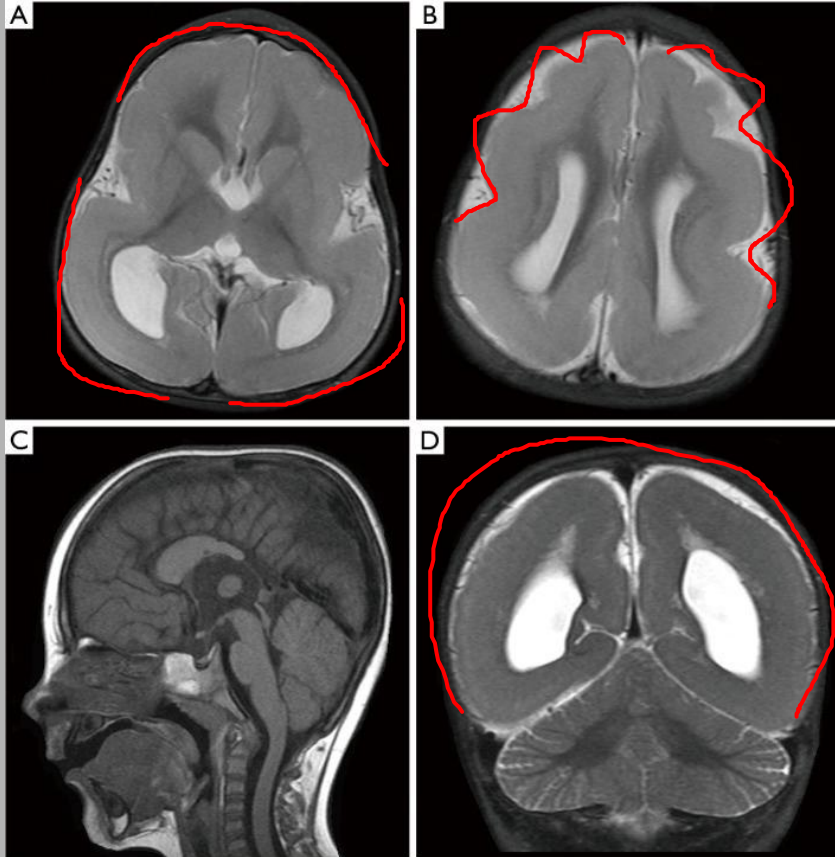
Likely also true in humans with molecular layer heterotopia.



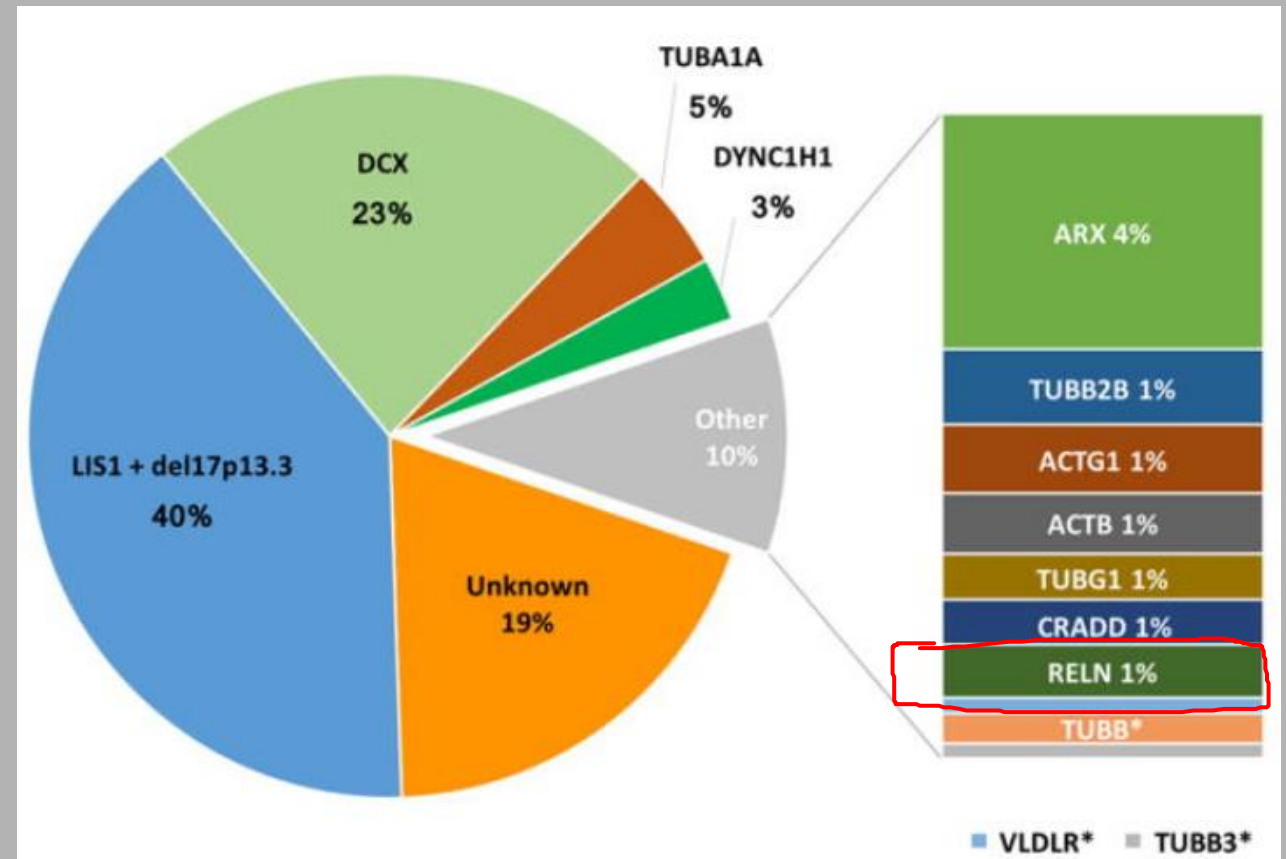
data from Allen Brain Atlas



Neuronal placement disorders: *Lissencephaly – smooth brain*



Tan AP et al. 2018 Quant Imaging Med Surg
<https://www.ncbi.nlm.nih.gov/pubmed/30211035>



DiDonato et al. 2018
<https://www.ncbi.nlm.nih.gov/pubmed/29671837>

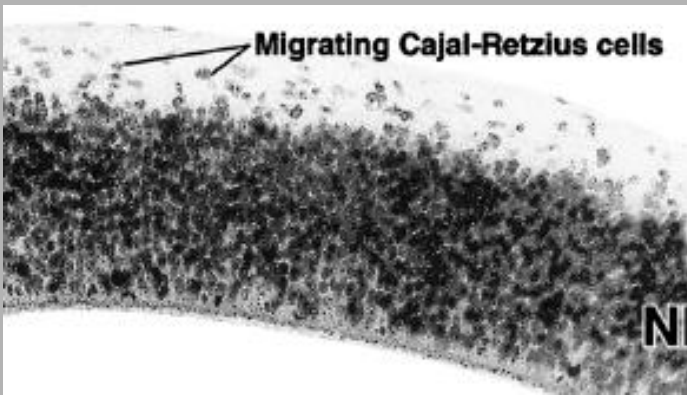
* **Agyria**: no gyri

* **Pachygyria**: few, large, broad gyri
-aka Incomplete lissencephaly

* **Lissencephaly**: smooth brain surface

Cortical thickness often greater than seen in normal individuals
Lack of apparent lamination of neurons.





Lissencephaly due to Reelin mutations:

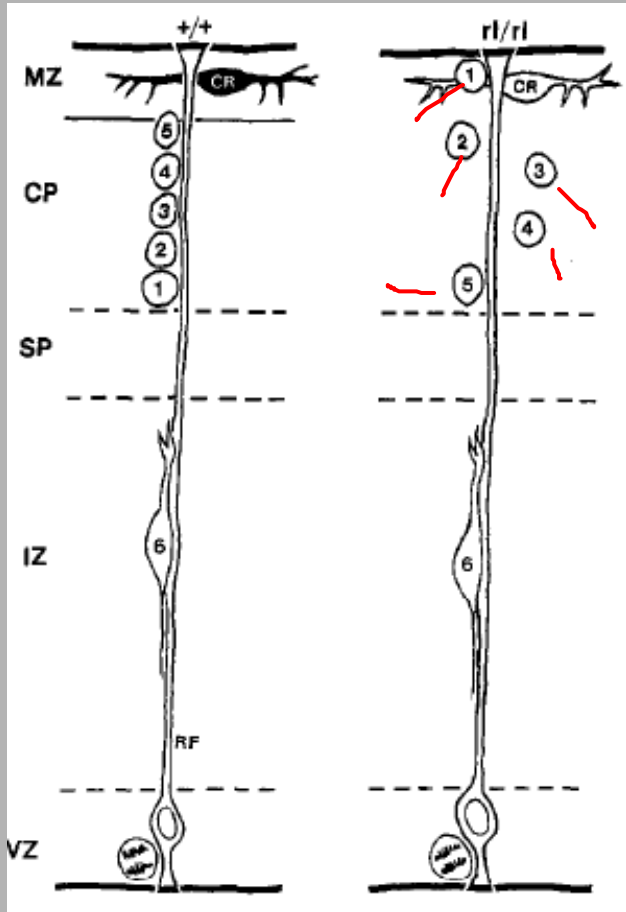
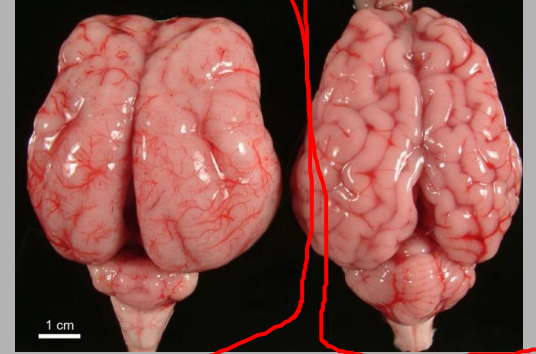
Cajal-Retzius cells are found in the marginal zone of the developing neocortex and secrete reelin. Reelin is an important molecular cue that guides neurons to their appropriate layer.

Sheep w mutant Reelin gene



Mutant

Normal

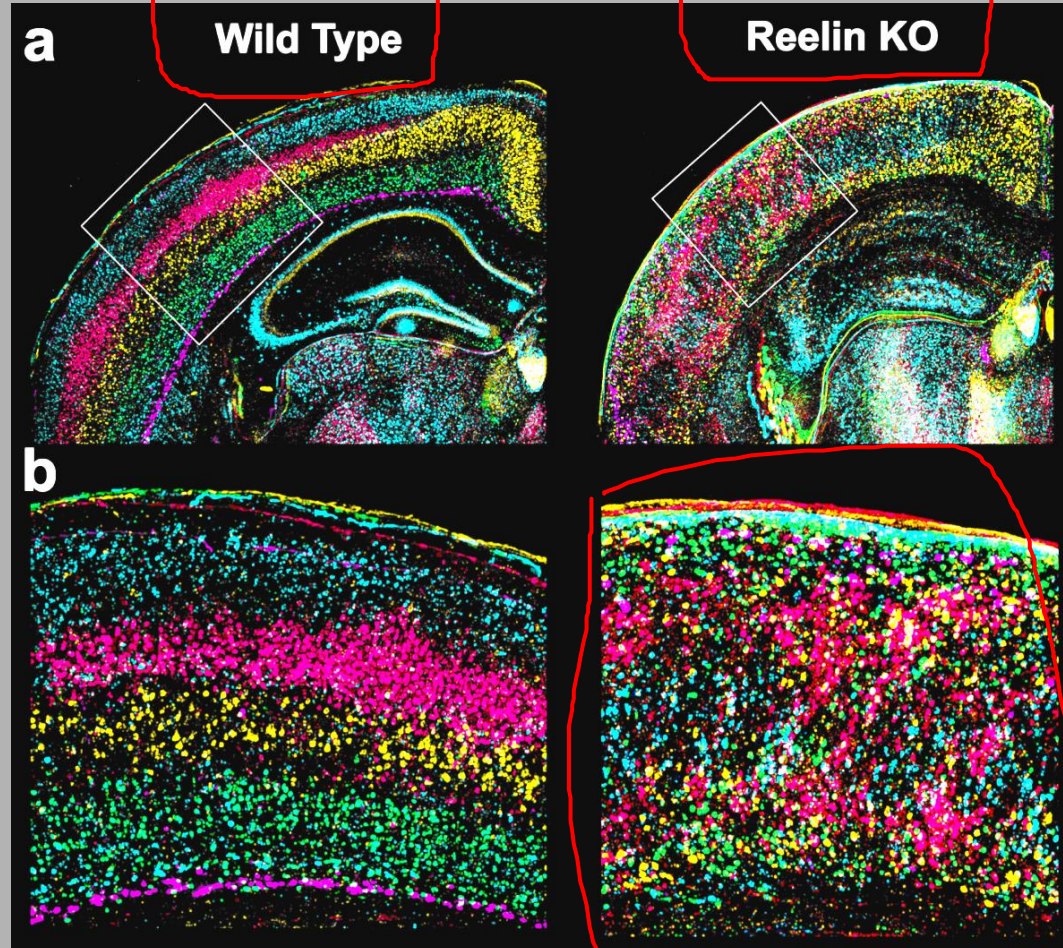


Rakic & Caviness 1995

Mouse with mutant Reelin gene

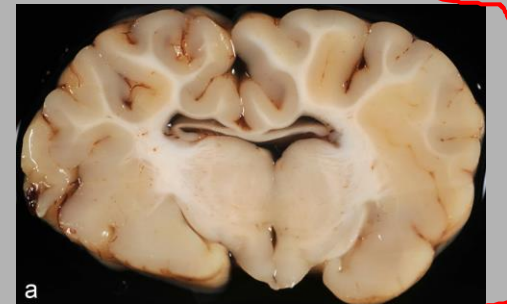
Wild Type

Reelin KO

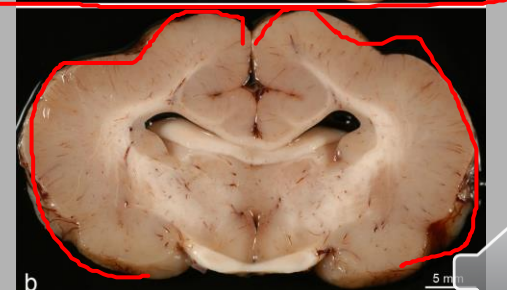


Boyle et al. 2011

Normal



Reelin mutant



Perez et al. 2013

Neuronal Migration & Placement

- Radial glia (RG) fibers contact the pial surface forming a substrate onto which newborn neurons can adhere & migrate in an inside-outside pattern.
- Migrating neurons must recognize the molecular cues that instruct the direction, distance, and destination of their journey. Intracellular proteins like microtubule associated proteins contribute to shape change and movement capabilities of neurons to reach their appropriate layer.
- Deficits in migration cause heterotopia and lissencephaly.



Please email with questions:

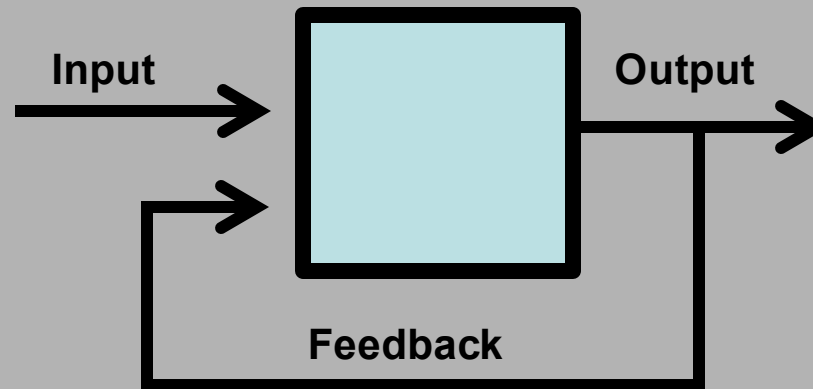
rramos02@nyit.edu

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



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